

Network Management and Monitoring

AI5049, Winter Semester 2026/27

12 October 2026

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About me

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- M.Sc. Applied Computer Science, research stay at the University of Toronto

What is this course about?

We work on real network problems and the evidence needed to solve them.

- Find what is actually broken
- Make changes without making things worse
- Measure whether the service works
- Turn operational questions into science

Our lens: large IP networks

We look at networks that carry traffic between organizations and regions.

- ISP and carrier backbones
- transit, peering and IXPs
- interdomain routing
- automation and monitoring at scale

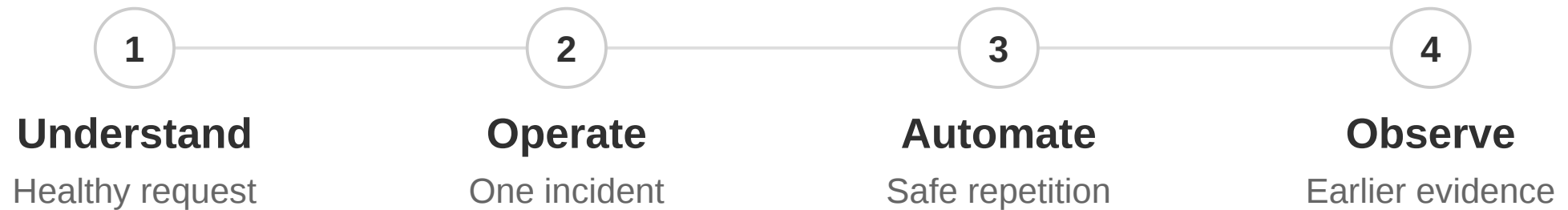
The same principles apply elsewhere. Our operating context is the Internet.

The technical block

Date	Time	Topic
12 October 2026	13:45–17:00	Networking 101 + Operations
13 October 2026	08:45–15:15	Automation + Incidents
14 October 2026	08:45–15:15	Monitoring / Observability

Incidents close day 2, after the Automation workflow.

One lab carries the technical block



The topology stays familiar. The question changes.
The slides remain the workshop guide.

The sessions after the lab

Friday sessions: **13:45–18:45**

Date	Focus
30 October 2026	Research questions and sources
13 November 2026	Experimental design
27 November 2026	Analysis and scientific argument
11 December 2026	Forecasts, control loops and agents
22 January 2027	In-class peer review and revision
5 February 2027	Presentations and reproducibility

Fridays: work on your project

The shared part gives you a method. Most of the session is project work.

- Short input together
- 45 minutes of shepherding with me per team
- Code, data, experiments or writing
- From November: brief team updates

Bring blockers early. We can fix scope before it becomes a deadline problem.

Your project

One network-management question, teams of 3-4.

- Build a system or analyze data
- Compare against a baseline
- Collect evidence for one limited claim

Paper: six pages including references. English, IEEE two-column.

Some project ideas

Area	Possible comparison
Monitoring	Polling vs streaming for short failures
Digital twins	Emulation vs modeling before deployment
Automation	Automated vs manual emergency repair
AI for networking	LLM troubleshooting vs a runbook

Bring your own question if you can evaluate it.

Research venues close to this course

Venue	Typical contribution
<u>IEEE/IFIP NOMS</u>	operating, managing and automating networks
<u>CNSM</u>	management systems, telemetry and orchestration
<u>ACM IMC</u>	measuring and understanding Internet behavior

Start here for literature on **management, monitoring and measurement**.

What counts towards your grade?

Individual work in the **paper and artifact** is assessed.

Paper area	Weight
Introduction & related work	20%
Approach & methodology	25%
Evaluation & results	30%
Discussion & conclusion	15%
Abstract, writing & reproducibility	10%

20 + 25 + 30 + 15 + 10 = 100%.

Who did what?

Record individual and joint work in `DECLARATION.md`.

Record	Make it concrete
Paper	Sections, figures and revisions
Artifact	Code, configuration and experiments
Joint work	Shared task and each person's contribution

Point to the work so the contribution can be checked.

What goes with the paper?

The paper makes the argument. The **artifact** lets me check it.

- Code and configurations
- Data and software versions
- Reproduction steps
- Seeds, failed runs and exclusions

Public repository: optional. Access for the examiner: required.

Scientific integrity

Follow the IEEE publishing ethics requirements.

- Cite ideas, methods, data and figures
- Check references against the original
- Report what you actually measured
- Keep counter-evidence
- State limitations and uncertainty

A result against your hypothesis is still a result.

AI use and disclosure

Follow the IEEE policy on AI-generated content.

- Spelling and grammar fixes only: disclosure optional
- Generated or substantially rewritten text: disclose
- Generated figures or code: disclose
- Name the system, affected parts and extent in the **Acknowledgments**

You are responsible for the references, code and technical claims.

Dates and deadlines

Date	Deliverable / session
11 January 2027, 23:59	Full draft + supporting material → Moodle
22 January 2027, in class	Read, write and hand in the paper review
5 February 2027	Presentations in class (ungraded)
12 February 2027, 23:59	Paper, artifact, declaration → Moodle

Optional peer-review bonus

One qualifying review improves the individual grade by **one step**.

Example: **2.3** → **2.0**.

The best possible grade is **1.0**. Failing grades remain unchanged.

The review should fit on **one page**.

Read in class. Write **legibly by hand**. German or English accepted.

What makes a qualifying review?

A qualifying review covers four things:

- **Summary:** question and main result
- **Method:** assess a methodological choice
- **Judgment:** strength and limitation with reasons
- **Revision:** propose a feasible improvement

I record the four criteria and my reasons.

Where are we starting?

Five minutes. Work alone first, then compare in pairs.

Topic	Used before	Can explain	New to me
IP, routing and DNS			
Linux and logs			
Python, APIs and Git			
Papers and experiments			

Add one thing you want to get better at.

Administration ends here.

From now on, **one network carries the technical block.**

The concepts travel further than the lab.

One network. Four questions.

Module	The question
Networking 101	How does this network actually work?
Operations	What broke, and how do we restore it safely?
Automation	How do we repeat that change safely?
Observability	How could we detect and explain it first?

understand → operate → automate → observe

Then challenge the evidence.

observe → question → measure → research

Research tests what our observations can establish.

The project turns that evidence into a paper and artifact.

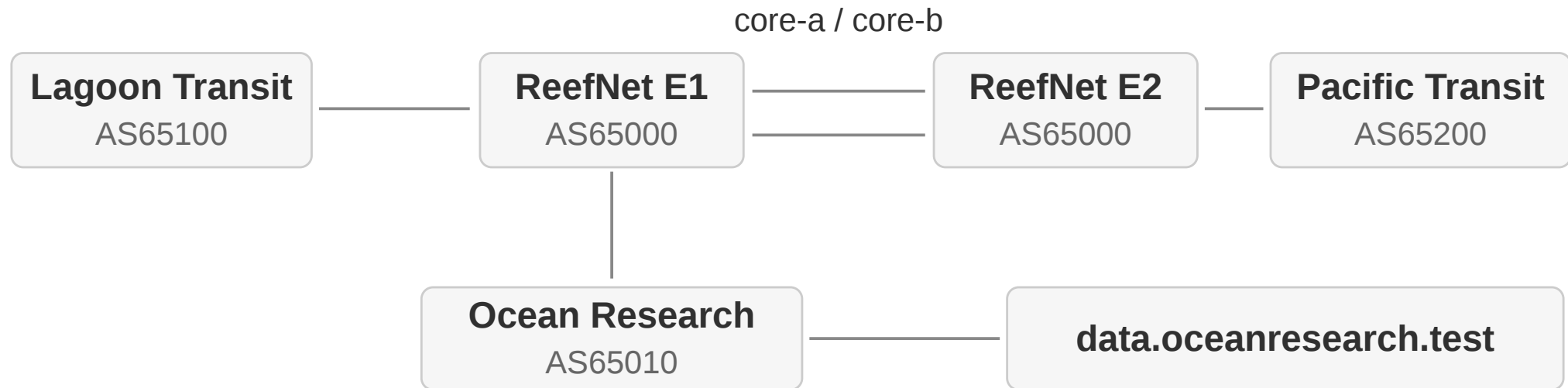
Peer review tests the argument.

Meet ReefNet at BOB1

ReefNet operates BOB1 on Bora Bora.

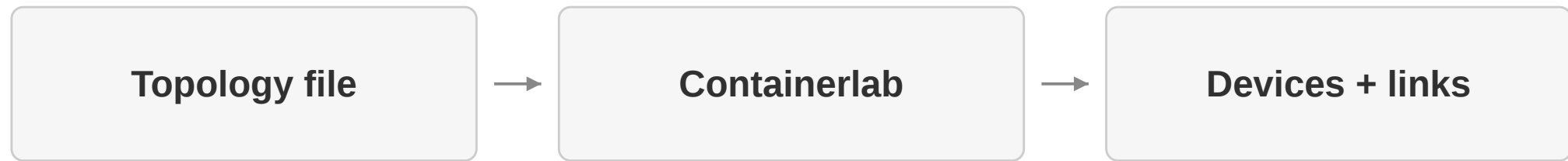
Bora Bora Ocean Research needs its data service reachable.

Lagoon Transit and **Pacific Transit** are our upstreams.



Containerlab creates the network.

It creates the topology, virtual devices and links we operate.



Creation is not verification.

We will recreate the same baseline at each technical session.

NetBox is not the router.

NetBox describes what we intend to exist.

Inventory connects devices, interfaces, circuits and addresses.
Follow those relationships before choosing a change target.

In BOB1, trace the Lagoon circuit to its ReefNet interface.

An inventory record cannot tell you whether packets pass now.

The router shows the present.

The CLI shows what is **configured and operational now**.

Configured	Operational
Interface enabled	Carrier up or down
Export policy attached	Prefix advertised or absent

Keep the observation time. “Now” does not stay current.

Grafana shows observations over time.

Check the age and scope of the data behind a panel.



A panel is the end of an observation pipeline.

A green panel can contain old data.

Look from outside.

An external probe shows the service from **another vantage point**.

Lagoon and Pacific probes can see different outcomes.
Neither sees every possible customer path.

The customer is not your monitoring system.

Four views. Four questions.

View	Question
NetBox	What should exist?
Router / CLI	What exists now?
Grafana	What has been happening over time?
External probe	What does the service look like from outside?

Disagreement is evidence. It is not a reason to trust the prettiest screen.

Prepare the lab once.

Before the first session, open PowerShell: `ws1 -d Containerlab`

Run the following commands **inside WSL**:

```
cd ~  
git clone https://github.com/Stinktopf/network-management-and-monitoring.git  
cd network-management-and-monitoring  
make setup
```

Wait for **READY FOR CLASS**. All four scenarios are checked.

The runtime stops. NetBox stays prepared. Diagnostics: `.state/`.

Know the three reset levels

Command	Use it when	What it keeps
<code>make reset</code>	you changed or broke the current exercise	prepared NetBox data
<code>make down</code>	you are finished for now	NetBox data and downloaded images
<code>make clean</code>	you want a complete AI5049 factory reset	downloaded base images only

After `make clean`, run:

```
make setup
```

`make clean` is **project-local**. It does not run a global Docker prune.

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Networking 101

How does this network actually work?

Follow **one service request**, then its response.

Open the lab

In the **lab-directory terminal**, start the healthy network:

```
make networking
```

Optional: run `code .` here to open the lab folder in VS Code.
The Containerlab extension opens `lab.clab.yml` in TopoViewer.

Then enter the Lagoon probe:

```
make enter NODE=probe01.bob1.lagoontransit.test
```

Follow one service request.

In ReefNet: a Lagoon probe requests `data.oceanresearch.test`.

One request crosses several independently operated links.



What must work before the application can return a response?

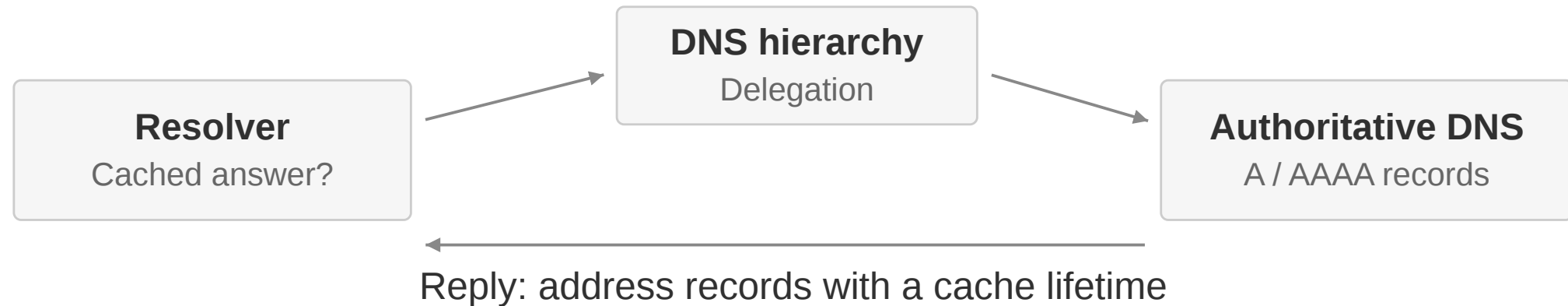
APPLICATION → TRANSPORT → IP → ROUTING → LINK → PATH

The TCP/IP stack

Application	HTTP, DNS
Transport	TCP, UDP
Internet	IP, ICMP
Link	Ethernet
Physical	Copper, fiber, radio

Each layer carries data for the layer above it.

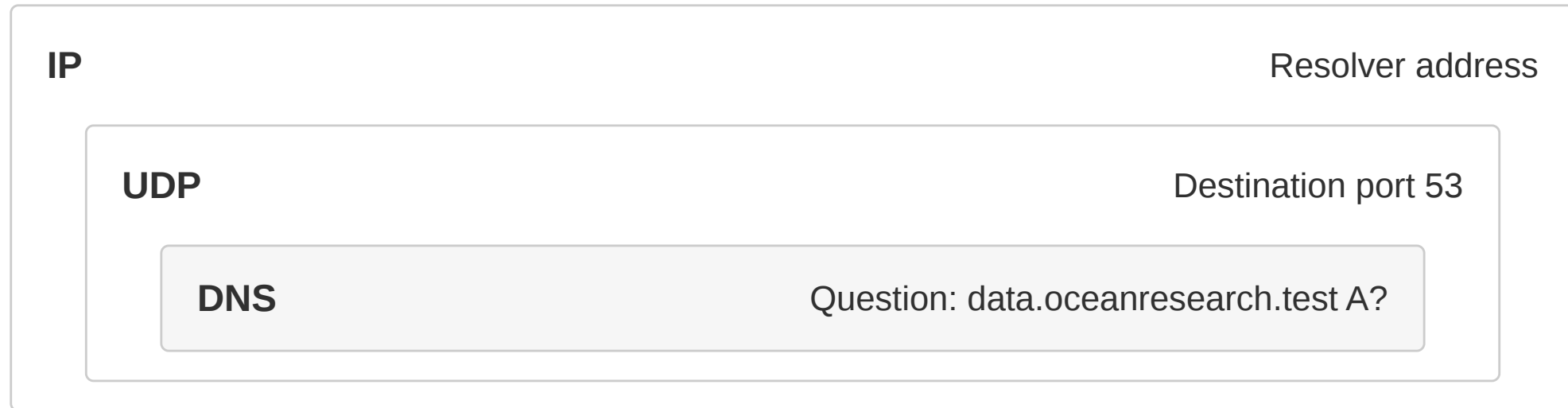
A name needs an address.



An address is an answer about naming, not service health.

A records carry IPv4 addresses. AAAA records carry IPv6 addresses.
ReefNet probes query the lab DNS server directly.

A DNS query becomes transport data



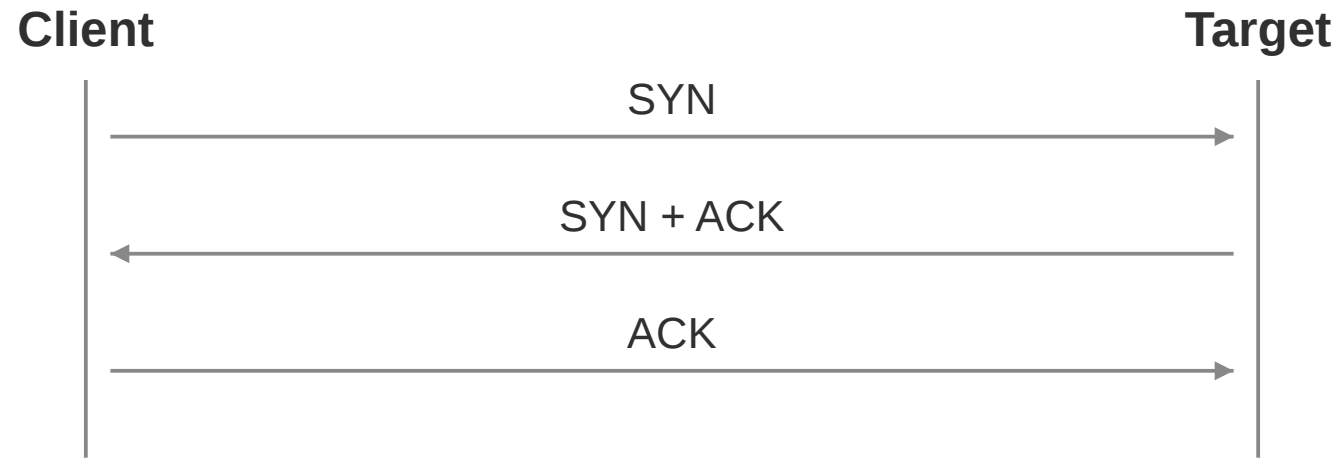
This query uses UDP. DNS also supports TCP.

TCP and UDP

	TCP	UDP
Data	Ordered byte stream	Individual datagrams
Connection	Required	None
Retransmission	Built in	Application decides

Both use IP addresses and ports. The application chooses the transport.

TCP gets the conversation started.



TCP setup precedes the application response.

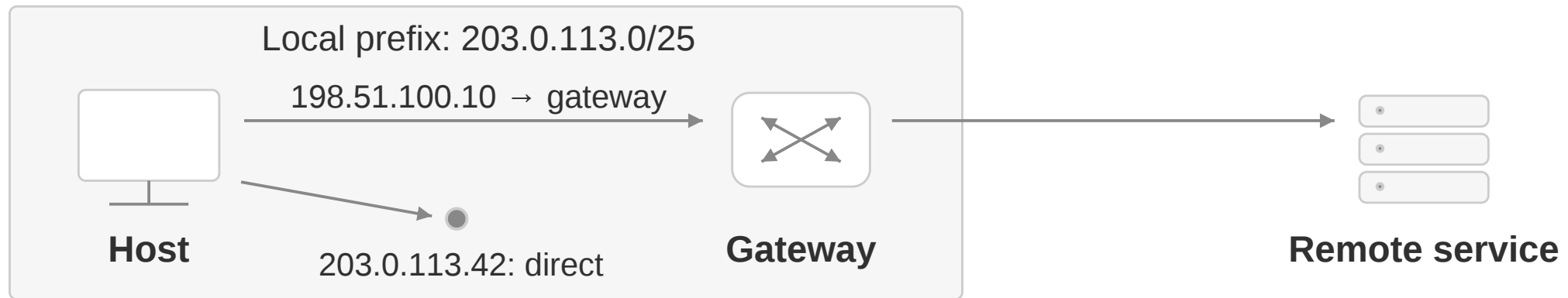
After DNS, our HTTP request opens a TCP connection.
TCP orders bytes, retransmits loss and controls congestion.
A completed handshake is not an application response.

IP gives the packet a destination.



IPv4 uses 32-bit addresses. IPv6 uses 128-bit addresses.
Routers choose a path from the destination IP address.

Is the destination local or remote?



A local destination is reached directly. A remote one needs a router.
Use `ip route` and `ip -6 route` to inspect the host's decision.

The route selects a next hop.



The packet keeps its final IP destination.

Resolve the next hop on the selected outgoing link.

The most specific route wins

A `/25` matches the first 25 address bits. Illustrative lookup table:

Prefix	Next hop
<code>0.0.0.0/0</code>	<code>192.0.2.1</code>
<code>198.51.100.0/24</code>	<code>192.0.2.2</code>
<code>198.51.100.128/25</code>	<code>192.0.2.3</code>

Destination: `198.51.100.200`

The `/25` is the **longest matching prefix**.

Forwarding chooses from routes that are already installed.

Now we need Layer 2

IP chooses the **next hop**.

On Ethernet, send a frame to that next hop's **MAC address**.

Neighbor discovery connects these two decisions.

MAC addresses belong to a link.

An Ethernet frame carries source and destination MAC addresses.

On the Lagoon probe:

```
ip -br link
```

MAC addresses are normally **48 bits** and identify local-link interfaces.

`ff:ff:ff:ff:ff:ff` is the Ethernet broadcast address.

For a remote IP destination, use the gateway's MAC address.

Ethernet carries frames across one link



A frame belongs to one link. Its FCS detects corruption.

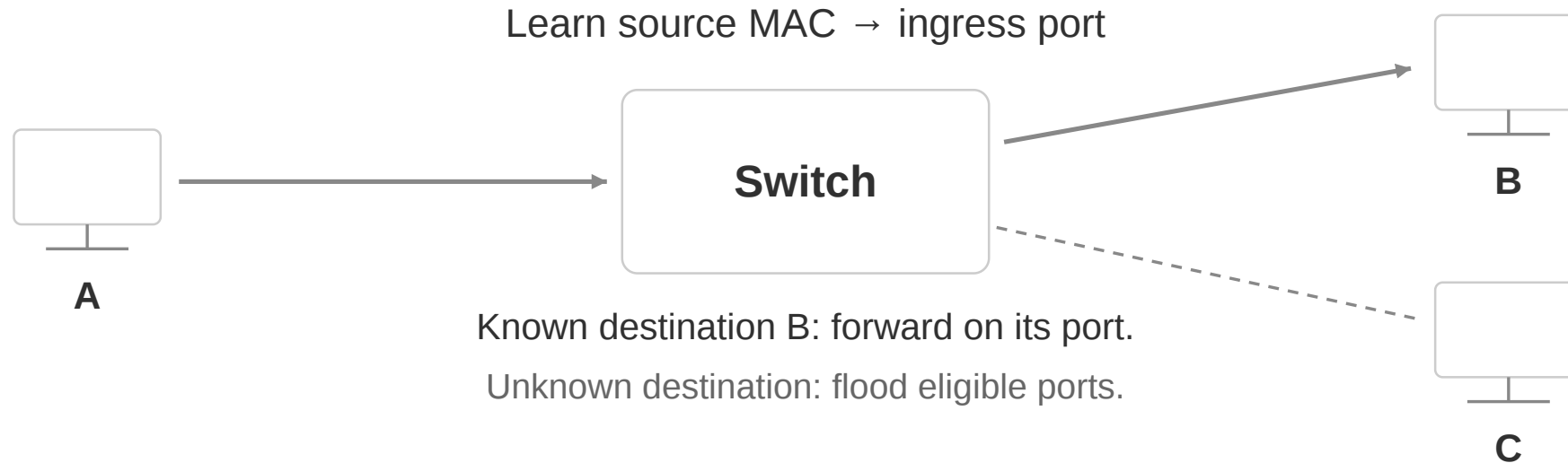
Bits need a physical link.

Medium	Signal	Fault evidence
Copper	Electrical	Link loss, errors, cable damage
Fiber	Light	Loss of light, low receive power

A route cannot repair a dirty connector.

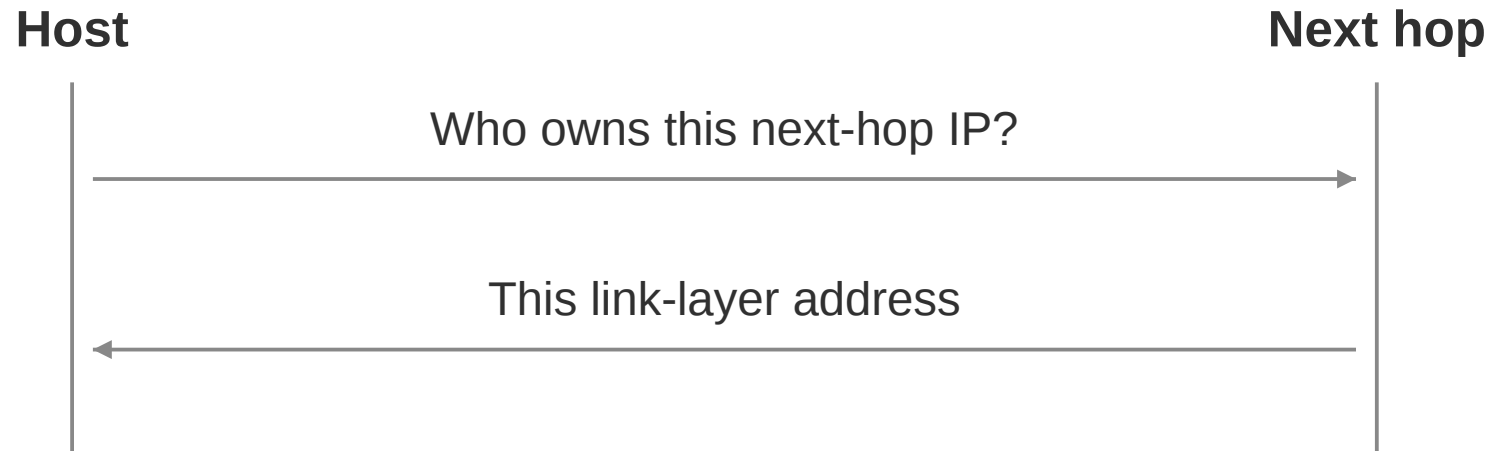
BOB1 has virtual links. Optical diagnostics need physical equipment.

A switch forwards frames by MAC address



Broadcast and multicast handling depend on the link and switch configuration.

Resolve the next hop.

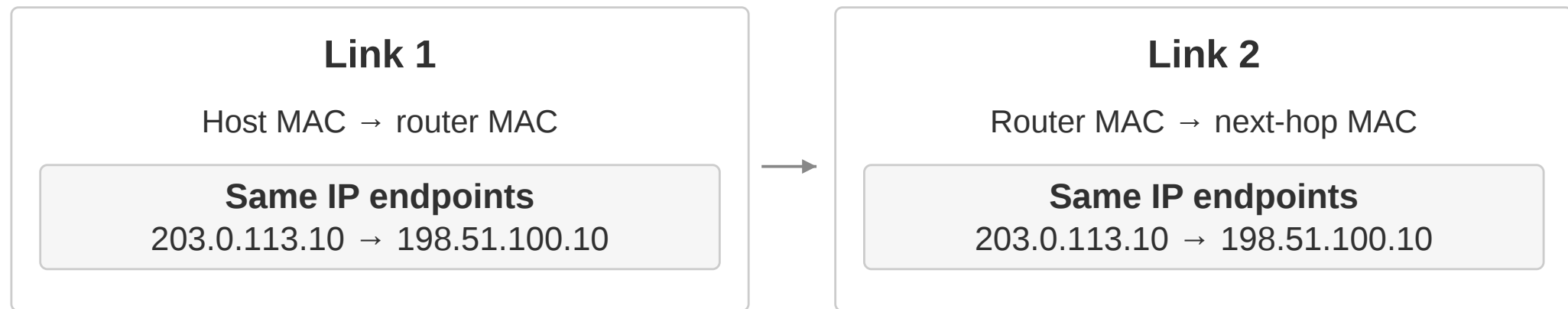


ARP uses broadcast. ND uses multicast solicitation.

IPv4 uses **ARP**. IPv6 uses **Neighbor Discovery**.

Resolve the next hop on this link.

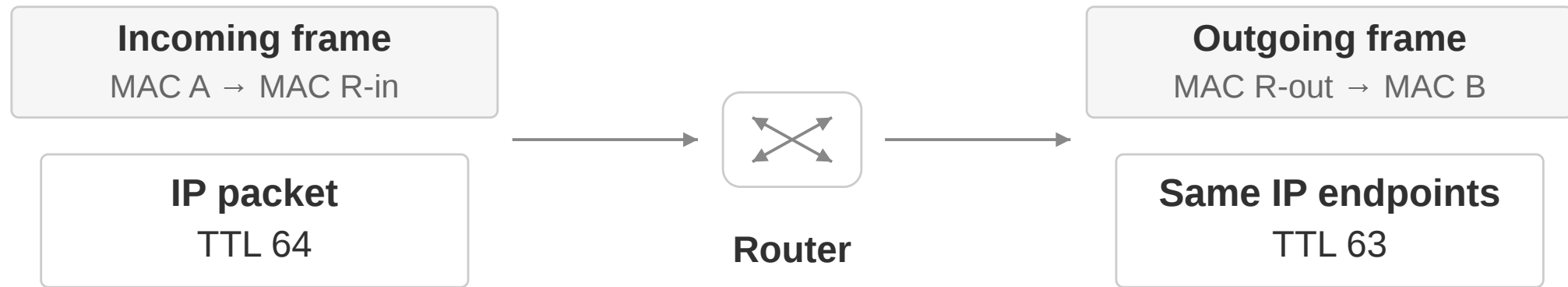
Remote IP, local MAC



Each router builds a new link-layer frame.

IP identifies the end-to-end destination. MAC addresses serve the current link.

Routers repeat the process hop by hop



Rebuild the frame. Keep the IP destination. TTL values are illustrative.

Each hop performs a new route lookup and builds a new frame.
TTL / Hop Limit is reduced so packets cannot loop forever.

Look inside the provider.

Keep the Lagoon probe open. In another lab-directory terminal:

```
make enter NODE=operations01.bob1.reefnet.test
```

Inside the operator workstation:

```
ssh edge01.bob1.reefnet.test
```

Use router FQDNs. The lab SSH configuration is already present.
Credentials: `admin / NokiaSr11!`. Leave the router with `quit`.

Where do routes come from?

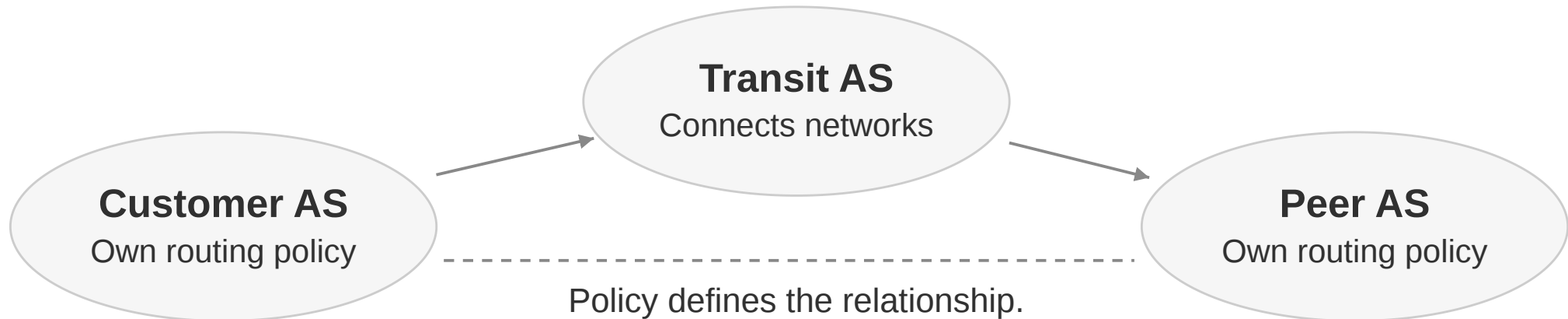
Route source	Origin
Connected	Addressed interface
Static	Operator configuration
Dynamic	Routing protocol

On ReefNet Edge 01:

```
show network-instance default route-table ipv4-unicast
```

Next: how IGP and BGP supply dynamic reachability.

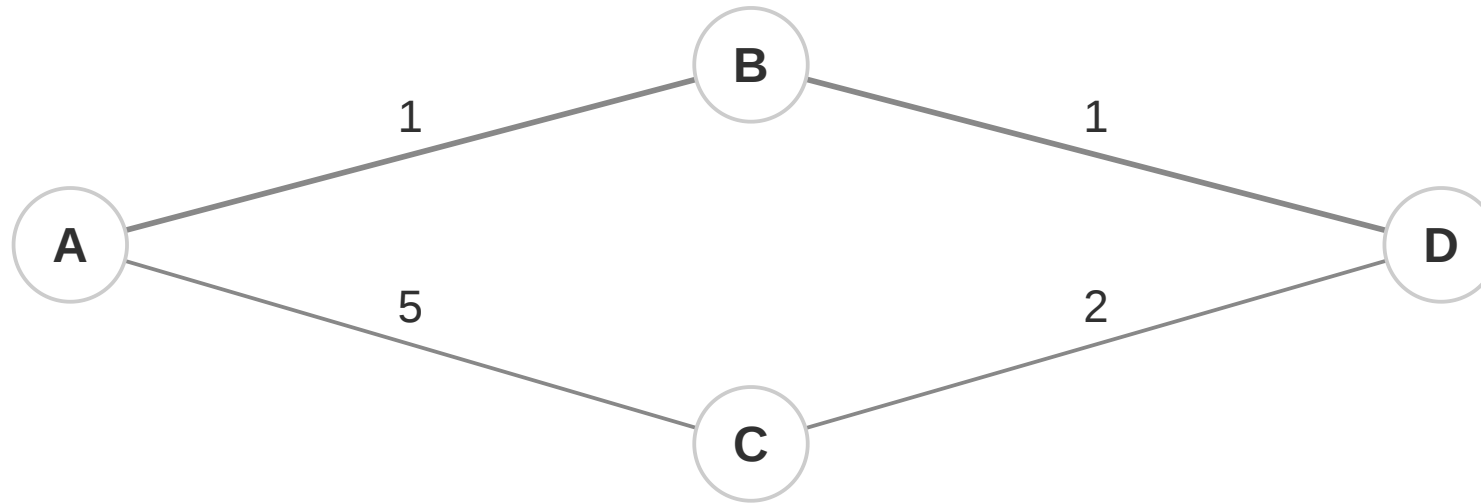
An AS is one routing domain



An AS is an administrative boundary, not one router.

AS: autonomous system, under one routing administration.
Inside: internal reachability. Between ASes: prefixes and policy.

Inside an AS: IGPs



Example link costs: the shortest internal path costs 2.

OSPF and IS-IS compute internal paths from link state.

In ReefNet: `show network-instance default protocols ospf neighbor`

BGP exchanges reachability.

BGP advertises IP prefixes together with path attributes.

Our lab has four autonomous systems.

On `edge01.bob1.reefnet.test`:

```
show network-instance default protocols bgp neighbor  
show network-instance default protocols bgp routes ipv4
```

A router can receive several routes to the same prefix.

BGP selects according to local policy and route attributes, not an IGP-style shortest path.

BGP is policy first.

For comparable routes to the same prefix:

1. apply local policy
2. prefer higher local preference
3. compare AS_PATH and further tie-breakers

Inspect the live prefix on ReefNet Edge 01:

```
show network-instance default protocols bgp routes ipv4 prefix 198.51.100.0/24
```

A longer path can be selected on purpose.

IGP and BGP

	OSPF / IS-IS	BGP
Carries	Link-state topology	Prefixes and attributes
Chooses by	Internal metric	Policy and attributes
Provides	Internal reachability	Route distribution

eBGP connects ASes. **iBGP** distributes routes inside an AS.
BGP next hops still need an internal path.

Transit and peering

	Transit	Peering
Reachability	Wider Internet	Peer and its customers
Payment	Customer pays provider	Often settlement-free

An **IXP** provides shared infrastructure for peering.
Business relationships shape route preference and export.

Who assigns the prefixes we route?



Administrative delegation, not a forwarding path

Delegation assigns address space. Routing advertises reachability.
The two are related, but they are not the same operation.

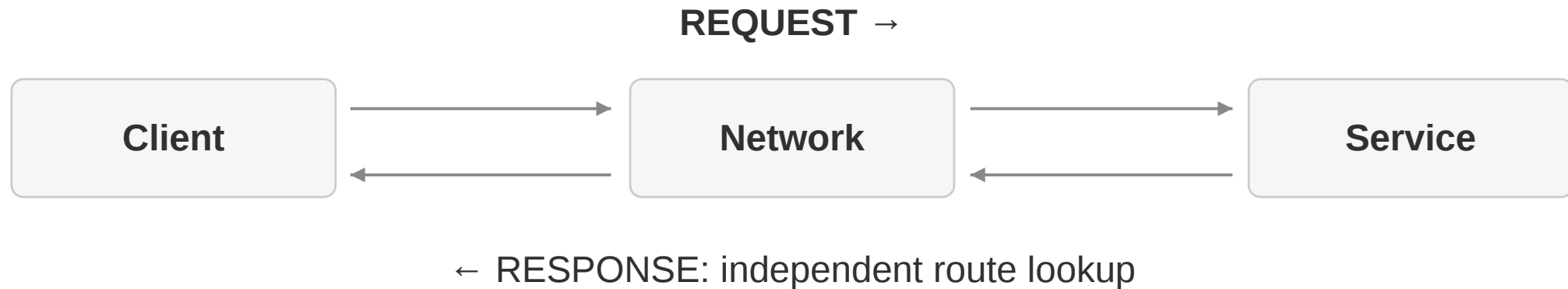
BOB1 uses documentation prefixes. Its service is not on the public Internet.

The destination still has work to do.



The application must accept the request and return a response.

The response needs its own route.



The destination now becomes the source.

Its routing decision is independent of the request path.

Reachability needs both directions. Paths may be asymmetric.

ICMP: errors and diagnostics

Message	Meaning
Echo Request / Reply	Reachability test
Destination Unreachable	Cannot deliver
Time Exceeded	TTL / Hop Limit expired
Packet Too Big (IPv6)	Packet exceeds link MTU

ICMP belongs to the Internet layer.

Ping uses ICMP Echo

`ping` sends an ICMP Echo Request.

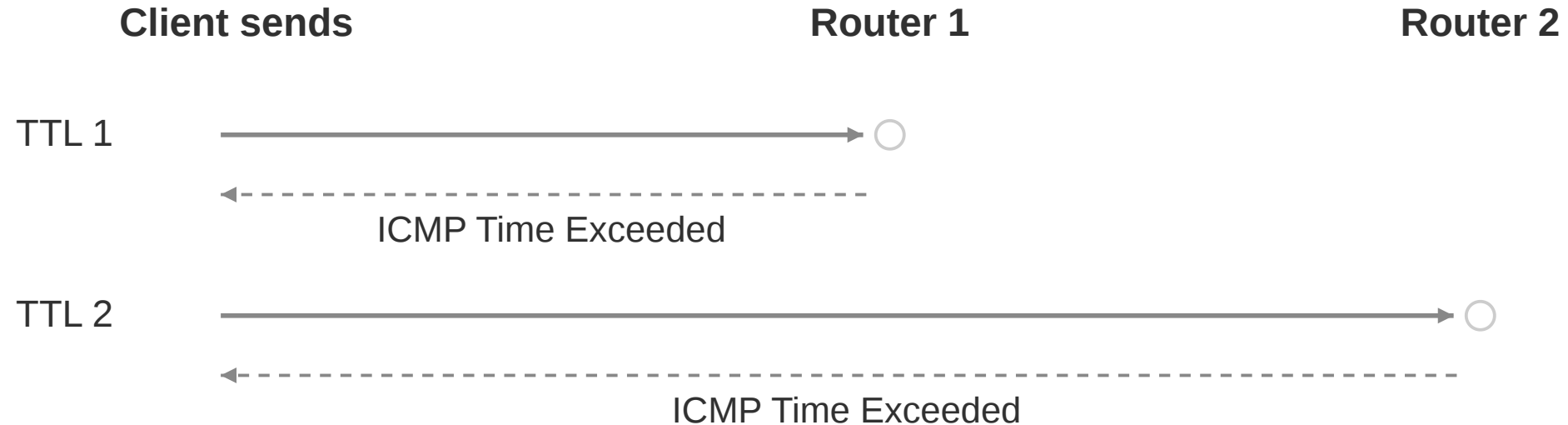
On `probe01.bob1.lagoontransit.test`:

```
ping -4 -c 3 198.51.100.10  
ping -6 -c 3 2001:db8:100::10
```

A reply shows that this ICMP exchange worked in both directions.

No reply does not identify where it failed.

Traceroute expires the TTL.



Successively larger TTL values reveal responding hops.
A missing reply does not identify the failed forwarding hop.

Read the live host from the bottom up

Stay on `probe01.bob1.lagoontransit.test`.

```
ip -br link
ip -br addr
ip route
ip -6 route
ip route get 198.51.100.10
ip neigh
```

For each command, ask one question:

What state did I just inspect?

Then test the path and protocol

Still on `probe01.bob1.lagoontransit.test`:

```
dig data.oceanresearch.test
ping -4 -c 2 198.51.100.10
traceroute -n -4 198.51.100.10
ss -lunpt
tcpdump -ni eth1
```

`ss` shows local sockets. Generate capture traffic from another terminal.
Stop `tcpdump` with **Ctrl+C**.

What does this mean in ReefNet?

Build one evidence chain for `data.oceanresearch.test`:

Evidence	Question answered
DNS response	Which destination?
Route + neighbor table	Which next hop and frame?
External HTTP request	Does the service answer?

On the Lagoon probe, finish with HTTP over both families:

```
curl --fail --max-time 5 -4 http://data.oceanresearch.test/  
curl --fail --max-time 5 -6 http://data.oceanresearch.test/
```

The service does not load.

The request crossed many boundaries.
Any one of them can fail.

Now stop reciting the stack and start collecting evidence.

A customer reports partial reachability.

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Network Operations

The service is broken.

What happened, and how do we restore it safely?

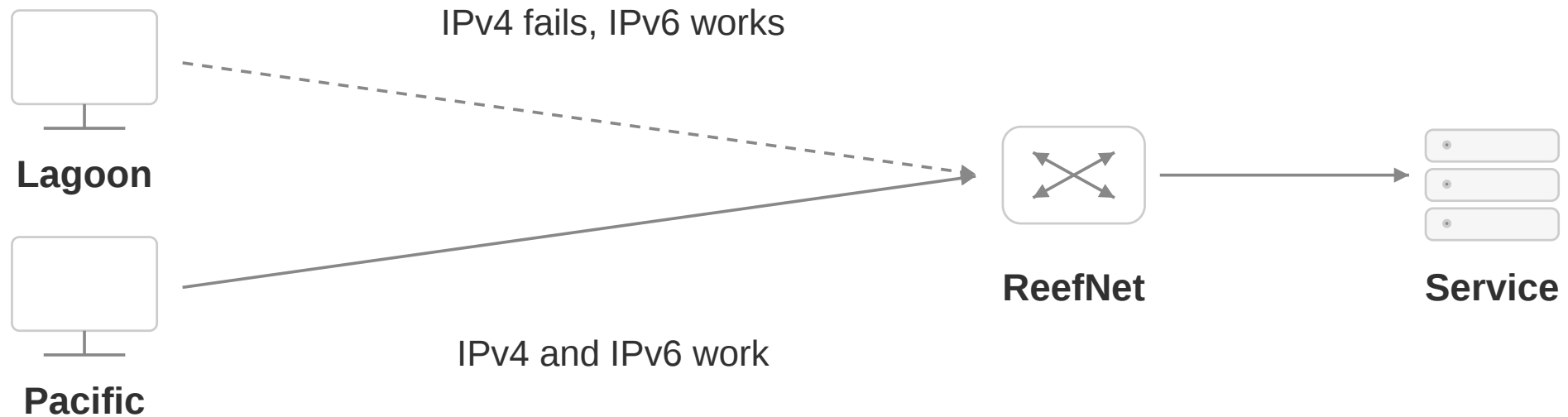
One incident, from report to recovery.



Evidence before action. Recovery before closure.

Establish known-good state before reproducing the failure.
Close the incident only when service and baseline agree.

A customer reports partial reachability



Record the source, address family, protocol and time.

REPRODUCE → BOUND → OBSERVE → HYPOTHESIZE → REPAIR → VERIFY

Establish the reference first.

Record a timestamped baseline. In the **lab-directory** terminal:

```
make networking  
make test
```

These are ICMP checks. Enter each external probe and also test HTTP:

```
curl --fail --max-time 5 -4 http://198.51.100.10/  
curl --fail --max-time 5 -g -6 'http://[2001:db8:100::10]/'
```

On ReefNet Edge 01, also save the healthy policy attachment:

```
info from running network-instance default protocols bgp group lagoon-v4
```

Start the Operations scenario.

From the lab directory:

```
make operations  
make enter NODE=operations01.bob1.reefnet.test
```

Evidence	Expected start state
BGP sessions	Established
Lagoon IPv4	Broken
Lagoon IPv6 + Pacific IPv4/IPv6	Healthy

`make reset` rebuilds this incident state.

Use the operator workstation.

From the operator workstation, connect to **one router at a time**:

```
ssh edge01.bob1.reefnet.test  
ssh edge02.bob1.reefnet.test  
ssh edge01.bob1.oceanresearch.test  
ssh edge01.bob1.lagoontransit.test  
ssh edge01.bob1.pacifictransit.test
```

All five routers use the same SR Linux CLI.

Lab login: `admin` / `NokiaSr11!`

Use `quit` to return to the calling terminal.

Reproduce the complaint from two places

In two **lab-directory** terminals, enter the external probes. Terminal A:

```
make enter NODE=probe01.bob1.lagoontransit.test  
ping -4 -c 2 198.51.100.10  
ping -6 -c 2 2001:db8:100::10
```

Another terminal:

```
make enter NODE=probe01.bob1.pacifictransit.test  
ping -4 -c 2 198.51.100.10  
ping -6 -c 2 2001:db8:100::10
```

Which combination fails?

Bound the incident.

	IPv4	IPv6
Lagoon	FAIL -----	PASS -----
Pacific	PASS -----	PASS -----

Keep the service and test method fixed. Vary the viewpoint.

Separate state from service.

Control plane	Routes learned and selected
Forwarding plane	Entries used to move packets
Service behavior	The customer gets a response

A healthy control-plane session does not prove that the service works.

Is BGP actually established?

On `edge01.bob1.reefnet.test`:

```
show network-instance default protocols bgp neighbor
```

For IPv4, focus on these peers:

Neighbor	Role
<code>192.0.2.0</code>	Ocean Research
<code>192.0.2.2</code>	Lagoon Transit
<code>10.255.0.2</code>	ReefNet edge02 iBGP

What does **established** prove?

REPRODUCE → BOUND → OBSERVE → HYPOTHESIZE → REPAIR → VERIFY

Established is not reachable.

The session is up. The customer route may still be missing.

Did the customer route arrive?

On `edge01.bob1.reefnet.test`:

```
show network-instance default protocols bgp routes ipv4 prefix 198.51.100.0/24
```

Then inspect IPv6:

```
show network-instance default protocols bgp routes ipv6 prefix 2001:db8:100::/48
```

A route in BGP is not yet proof of forwarding.

Is IPv4 installed for forwarding?

```
show network-instance default route-table ipv4-unicast prefix 198.51.100.0/24
```

Look for:

active route → resolved next hop → outgoing interface

The forwarding entry toward the customer is present.

ReefNet also needs internal reachability

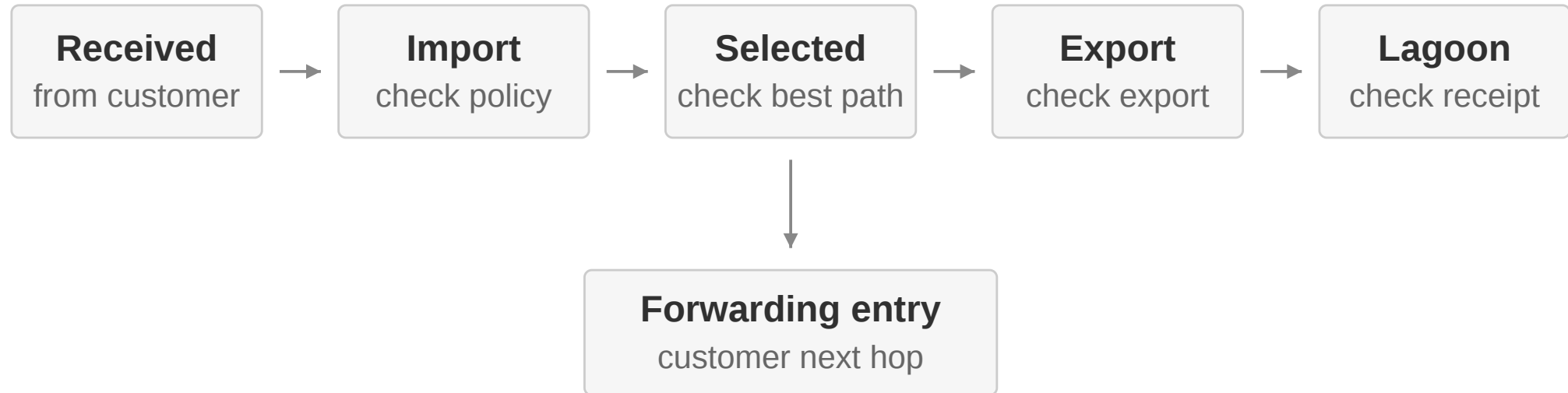
`edge01.bob1.reefnet.test` and `edge02.bob1.reefnet.test` use OSPF inside AS65000.

```
show network-instance default protocols ospf neighbor
```

The IGP gets us through **our** network.

BGP decides which external reachability we advertise and prefer.

The evidence now points outward



The route arrived and is installed. Next, compare the two upstream exports.

Compare the advertisements

On `edge01.bob1.reefnet.test`, toward `edge01.bob1.lagoontransit.test`:

```
show network-instance default protocols bgp neighbor 192.0.2.2 advertised-routes ipv4
```

On `edge02.bob1.reefnet.test`, toward `edge01.bob1.pacifictransit.test`:

```
show network-instance default protocols bgp neighbor 192.0.2.5 advertised-routes ipv4
```

Find `198.51.100.0/24`.

A testable hypothesis.



Predict the missing evidence before changing the router.

What evidence would disprove this hypothesis?

Inspect the policy attachment

On `edge01.bob1.reefnet.test`:

```
info from running network-instance default protocols bgp group lagoon-v4
```

Compare it with:

```
info from running network-instance default protocols bgp group lagoon-v6
```

What is attached only to IPv4?

Make the smallest justified change

On `edge01.bob1.reefnet.test`:

```
enter candidate  
  
delete / network-instance default protocols bgp group lagoon-v4 export-policy  
  
diff  
commit now
```

The group now inherits the healthy instance export policy.

Do not restart BGP. Do not touch `edge01.bob1.pacifictransit.test`.

The change is not the recovery proof.

Recheck the advertisement. From each external probe, test HTTP:

```
curl --fail --max-time 5 -4 http://198.51.100.10/  
curl --fail --max-time 5 -g -6 'http://[2001:db8:100::10]/'
```

In the **lab-directory** terminal:

```
make test
```

Fail on HTTP errors. Allow at most **5 seconds** per request.
Keep results from both upstreams and address families.

Prove success and containment.

Positive: the formerly failing Lagoon IPv4 service works.

Invariant: Pacific and IPv6 still work.

Negative: override absent. Unrelated configuration unchanged.

This repair tests policy attachment, not a new prefix allowlist.

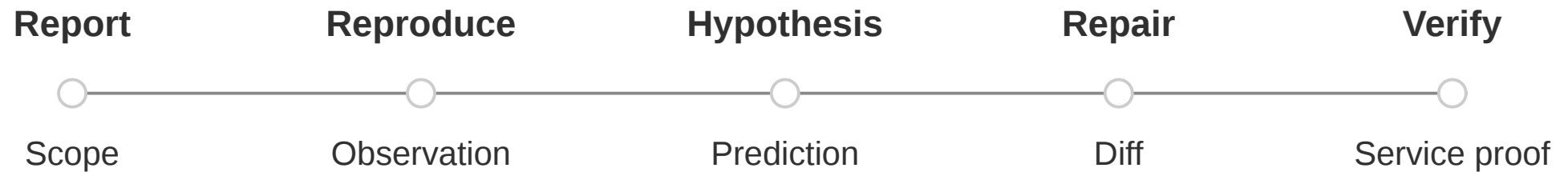
Confirm the known-good state.

After the service checks pass, compare configuration with the healthy baseline.

Explain any remaining difference. Keep the repair diff and service results.
A reachable service can still hide unintended changes.

Mark the state **known good** only when configuration and service agree.
Keep this evidence before resetting the lab.

Keep an incident timeline.



Record timestamps, actions and evidence at each boundary.

Separate observed facts, interpretations and actions.

Write the update from the evidence

A useful update states:

impact → **evidence** → **action** → **verification**

For example:

IPv4 through Lagoon Transit was missing because the customer route was filtered on export. The policy attachment was removed and reachability now succeeds from both external probes. IPv6 was unaffected.

Write the runbook

1. Reproduce and bound the symptom.
2. Compare intended and observed state.
3. Test a hypothesis before changing anything.
4. Repair, repeat the failing test, record the result.

Unexpected evidence means **revise the hypothesis**.

Write an RFO after the incident

RFO: Reason for Outage. Record impact, cause and recovery.

- **Impact:** IPv4 unreachable through Lagoon Transit
- **Cause:** wrong export policy attached to `lagoon-v4`
- **Recovery:** remove the attachment. Verify HTTP through both upstreams and families
- **Follow-up:** test IPv4 and IPv6 exports before deployment

Unknown time? Write **unknown**, not a guess.

RCA goes beyond the bad line

RCA: Root Cause Analysis. Look beyond the wrong attachment:

- Why did validation accept it?
- Why did the postcheck miss one address family?
- What test prevents the same failure next time?

Carry these prevention questions into the change contract.

That was network management.

Place this repair in the wider management picture:

- **Fault:** isolate lost reachability
- **Configuration:** repair export policy
- **Performance:** verify service behavior
- **Security:** control who may change the router
- **Accounting:** attribute resource use when needed

Configuration is only one part of operating a network.

Four views, one recovery claim.

View	Recovery evidence
NetBox	Intended service and circuit still match
CLI	Correct policy and route advertisements
Grafana	Fault and recovery history, if collected
External probes	Service works through both upstreams

No single view replaces the others.

Operations: before you leave

Save your timeline, repair diff and recovery proof.

```
make healthy  
make test  
make down
```

Tomorrow: make **this class of repair** safe to repeat.

13 October 2026

Automation + Incidents

Yesterday we repaired the network manually.

Today we need to make the same class of change **safely and repeatedly.**

At the end of today: **Who Broke the Internet?**

Same network, new job

From the lab directory:

```
make automation  
make enter NODE=operations01.bob1.reefnet.test
```

The same IPv4 export incident is back.

This time the repair needs a reviewed input and repeatable checks.

The safe change loop



Failed verification: record actual state, back out, verify again.

The workflow is the automation story.

YANG, NETCONF and gNMI are mechanisms within it.

INTENT → OBSERVE → PLAN → CHANGE → VERIFY → RECORD

Agree on a change contract.

Boundary	Agree before execution
Scope	Target, permitted diff, preserved state
Entry	Preconditions and reviewed input
Exit	Service checks and deadline
Recovery	Backout, owner and access path

If the contract is vague, the script automates the ambiguity.

The change contract

Target: **ReefNet Edge 01**, BGP group `lagoon-v4`.

	Required state
Before	Override: <code>BLOCK-CUSTOMER-V4</code>
After	Inherit <code>EXPORT-BGP</code> without a group override
Preserve	Lagoon IPv6 and Pacific Transit
Prove	Customer IPv4 export and external reachability

Which facts need a fresh read from the router?

INTENT → OBSERVE → PLAN → CHANGE → VERIFY → RECORD

Intent, policy, configuration

Level	ReefNet example
Intent	Customer reachable through both upstreams
Policy	Lagoon IPv4 inherits EXPORT-BGP
Configuration	lagoon-v4 currently has a group override

A correct policy on the wrong group still fails the intent.

Inventory gives the target context.

An inventory is a graph of relationships.



Join the intended circuit to the actual change target.

Use inventory to identify the target. Read its live policy before changing it.

Read the device inventory.

Inside the operator workstation:

```
TOKEN=$(cat /state/netbox-token)
NETBOX_API=http://netbox.bob1.reefnet.test:8080/api
curl -sG -H "Authorization: Bearer $TOKEN" \
  "$NETBOX_API/dcim/devices/" \
  --data-urlencode name=edge01.bob1.reefnet.test | jq .
```

The lab provisions the token. Keep it out of reports and screenshots.
Check that exactly the intended device is selected.

Read the circuit intent too

The same API exposes the Lagoon Transit circuit:

```
curl -sG -H "Authorization: Bearer $TOKEN" \
  "$NETBOX_API/circuits/circuits/" \
  --data-urlencode cid=BOB1-LAGOON-001 | jq .
```

Compare inventory intent with live state before changing the router.

NetBox describes intent. The router shows live state.

Join inventory into one change input

The API gives us separate objects. We need the relationship.

```
GET /api/dcim/devices/?name=edge01.bob1.reefnet.test
GET /api/dcim/interfaces/?device=edge01.bob1.reefnet.test
GET /api/ipam/prefixes/?prefix=198.51.100.0%2F24
GET /api/ipam/asns/?asn=65100
```

The useful part is not the endpoint itself.

It is joining the right objects into one approved change input.

Intent and observation can drift.

Compare **meaning**, not raw text.

Before calling it drift, check:

- same scope?
- same point in time?
- fresh observation?
- equivalent representation normalized?

Then decide whether to reconcile or investigate.

Unavailable input is not an empty list.

Input	State
Inventory snapshot	Current
Monitoring API	Unavailable
Previous output	Known working

Stop publication or use an approved, age-limited fallback.

Unknown targets must not silently disappear.

Freeze the input you reviewed

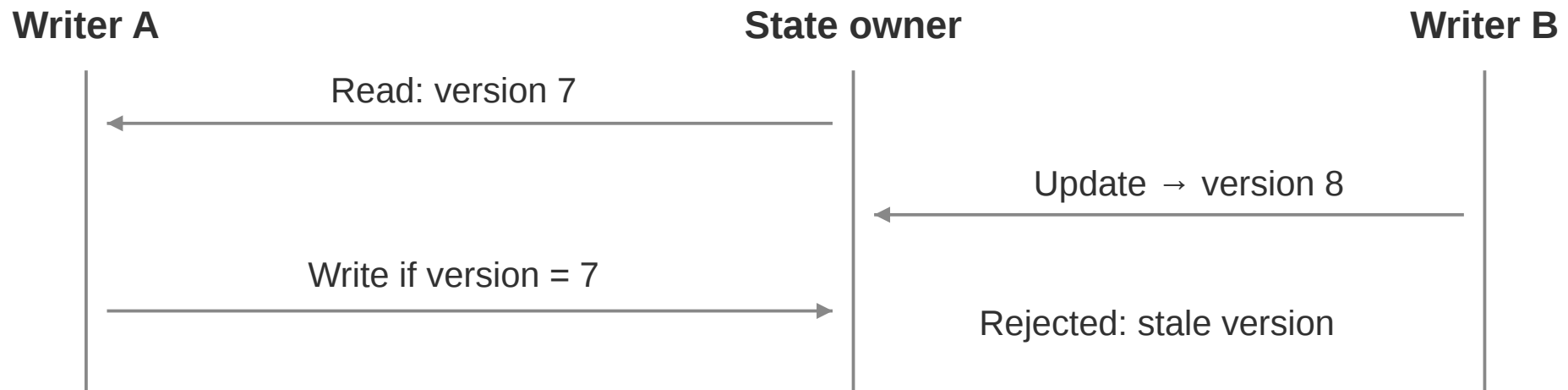
```
{  
  "change_id": "CHG-017",  
  "device": "edge01.bob1.reefnet.test",  
  "bgp_group": "lagoon-v4",  
  "expected_override": "BLOCK-CUSTOMER-V4",  
  "action": "delete group export-policy",  
  "expected_inherited_policy": "EXPORT-BGP"  
}
```

Reviewer and executor need the **same approved input**.

Also keep the current state needed for backout.

Prevent a lost update

Optimistic concurrency: write only if the version you reviewed is still current.



Check and write must be atomic at the state owner.

A CLI precheck is useful, but it cannot make the later write atomic.

Validate before rendering

The request is small. The failure modes are not.

Check	Catches
Device + BGP group	right action, wrong target
Current override	stale input or already repaired state
Direction	import edited instead of export
Inherited instance policy	deleting the override would expose the wrong fallback

Validate the request **and** the state it depends on.

Generate and inspect the candidate.

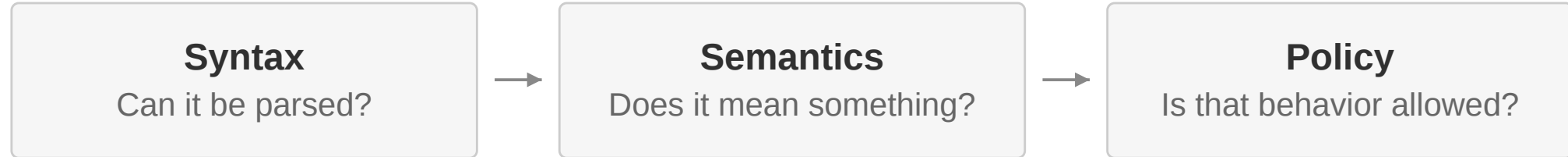
Generate the candidate from the frozen **CHG-017** input. Expected diff:

```
- export-policy [ BLOCK-CUSTOMER-V4 ]
```

Retain instance policy **EXPORT-BGP**. Reject unrelated changes.

Review now. Execute the lab repair later with gNMI.

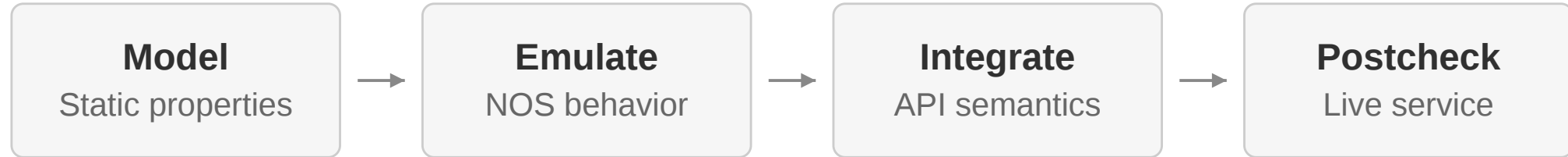
Validation has three layers.



A valid configuration can still violate the change contract.

Passing one gate does not prove the next.

Model it, emulate it, then check the service



Different tests expose different failure classes.

Passing a static model check cannot prove target API behavior.

Plan the maintenance window



Stop time + recovery owner apply at every gate.

Method of Procedure (MOP): steps, checks and recovery ownership

Agree on acceptable impact before the window opens.

Define success before the change

Check	This repair
Precondition	lagoon-v4 still uses BLOCK-CUSTOMER-V4
Postcondition	Customer IPv4 export and probes recover
Invariant	Lagoon IPv6 and Pacific remain healthy

Read routes locally. Test the service from outside.

Bound the rollout.

Choose a representative canary before increasing the target set.

1. Read live preconditions again.
2. Change one failure domain. Wait for service evidence.
3. Continue only while invariants hold.

Define **execution order**, **blast radius** and **stop time**.

Make the backout path reachable before the first write.

Give automation a bounded identity.

Identity	Read	Write
Telemetry collector	Required state	No
Change service	Preconditions + verification	Approved scope
Reviewer	Evidence	No

Load credentials at runtime. Record actor and change ID, never secrets.
Test emergency access and credential rotation before the window.

Execute and recover

Now assume the candidate is approved.

The next question is different:

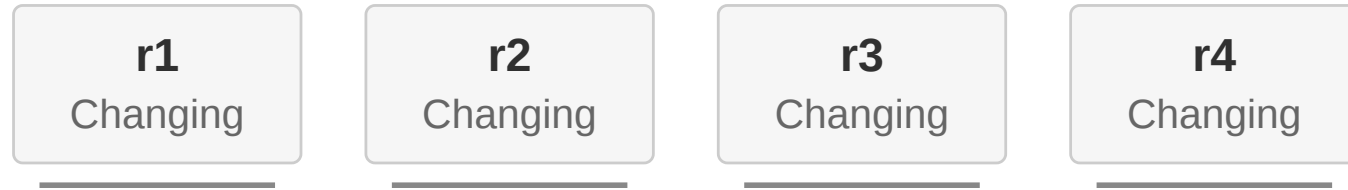
What happens when the write only partly succeeds, or we cannot tell?

A fleet changes the failure model.

One at a time



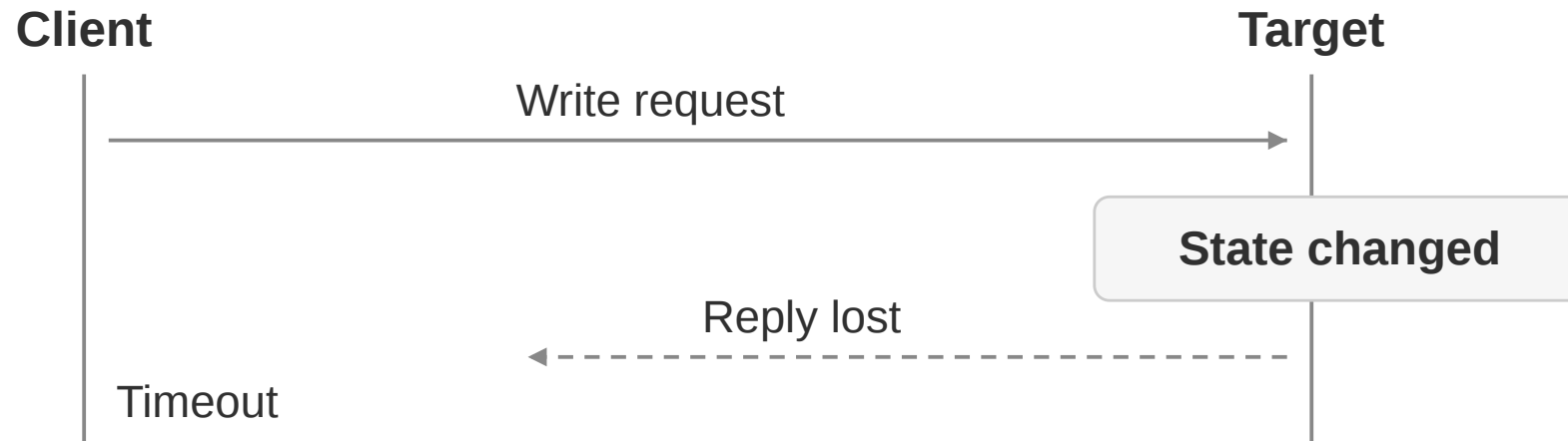
All at once



How many devices can this change affect before the next check?

More concurrency increases throughput and blast radius.

The write may already have happened



Unknown outcome $\neq$ no change. Read before retrying.

Record an unknown outcome explicitly in the audit trail.

Run it twice. What changes?

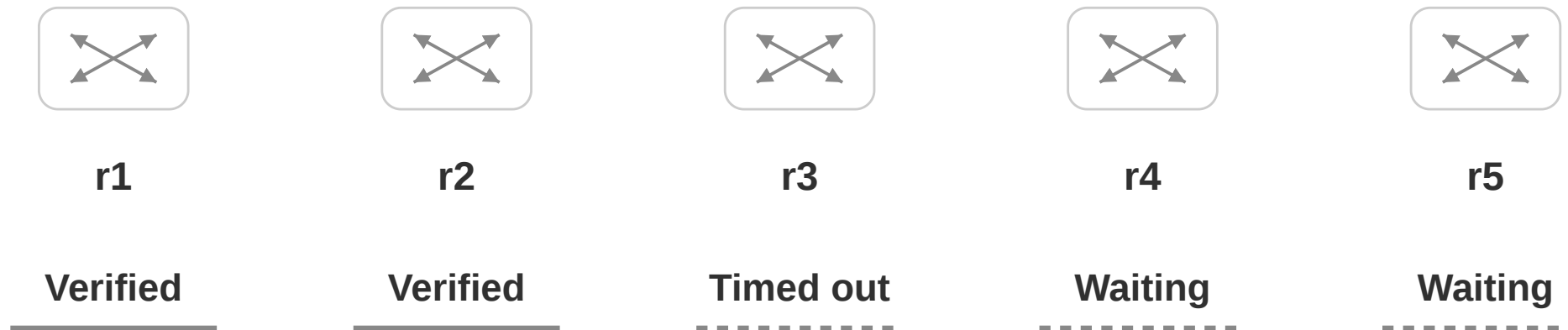
Idempotence, for a fixed operation F and state s :

$$F(F(s)) = F(s)$$

Useful after an uncertain outcome:

- repeated execution does not keep changing state
- retry is still not automatic
- service verification is still separate

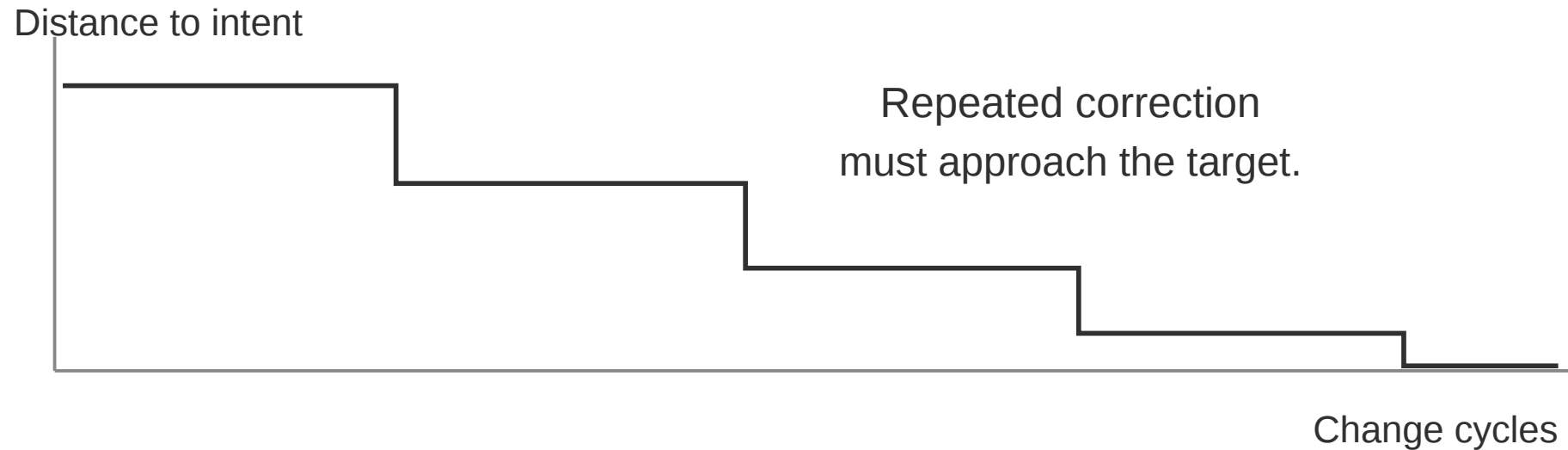
One router timed out after the write



Progress is per device. A timeout is not a rollback.

Which device needs a fresh read before any retry?

Does the loop converge?



Idempotence alone does not prove convergence.

Job success is not service success.

Exit code: 0

Probe: timeout

The script has successfully finished being wrong.

Test the intended behavior

The device accepted the change.

That only answers:

"Was the write accepted?"

Now test what should pass, what should fail, and what must remain unchanged.

Plan the repair verification

Check	Expected
Lagoon: 198.51.100.0/24	Export restored
Lagoon: 2001:db8:100::/48	Still exported
Pacific IPv4 path	Still healthy
Both probes, both families	Service responds

Use this matrix after the gNMI repair. The lab is still in the fault state.

Test rejected routes too.

For a different change, test route admission.

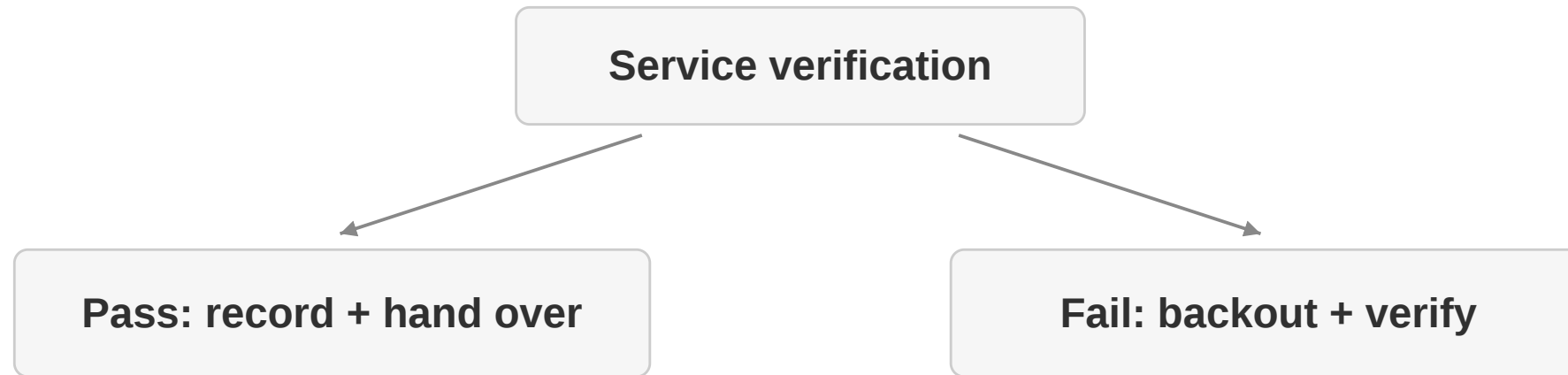
Route Origin Validation (ROV) checks RPKI origin authorizations.

Result	Meaning
Valid	Origin and prefix length permitted
Invalid	Covered, but no authorization permits this route
NotFound	No covering authorization

Policy decides the action. Test accepted and rejected routes.

This example is outside the BOB1 lab.

Rollback and backout



Rollback: configuration. Backout: recovery procedure.

Record the backup's identity, time and version.
Assign an owner to the full recovery procedure.

Which state are we restoring?

03:00. Someone says: "restore last Tuesday."

That is not precise enough.

Golden configuration	Backup
approved baseline	captured device state

Record identity, time, version and checksum.

Keep history outside the device failure domain.

Keep an audit trail.

```
{
  "actor": "course-automation",
  "target": "edge01.bob1.reefnet.test",
  "change_id": "CHG-017",
  "deleted_override": "BLOCK-CUSTOMER-V4",
  "inherited_policy": "EXPORT-BGP",
  "lagoon_ipv4_probe_passed": true
}
```

Keep inputs, timestamps and before/after evidence.

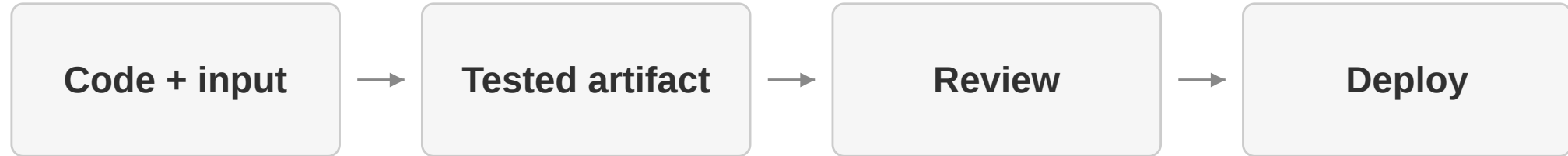
Never keep credentials in the audit record.

Who operates the automation?

- A named owner and bounded runtime
- One writer per target with explicit retry rules
- Fresh inputs and verified outcomes
- Tested recovery and a retained audit trail

The next shift must be able to run it.

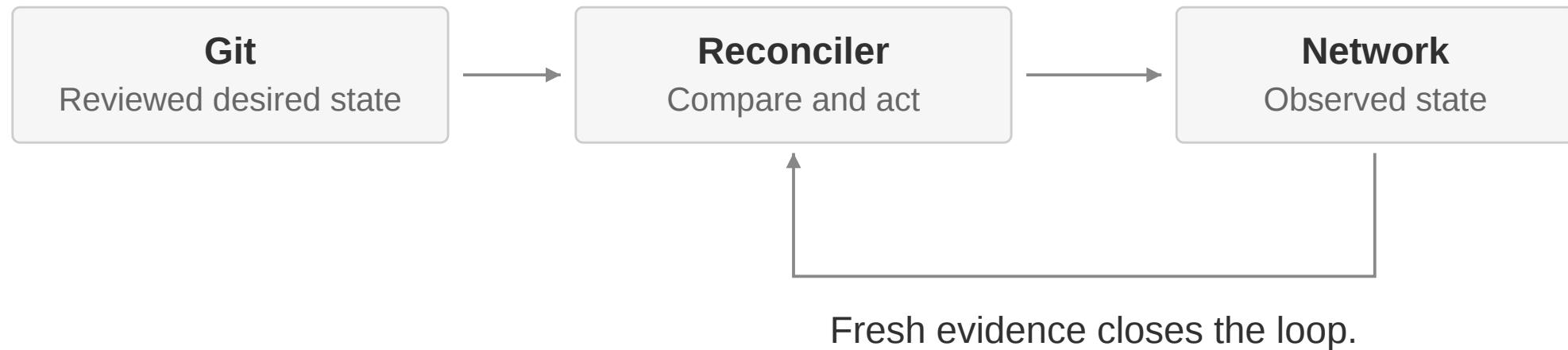
Promote the tested artifact



Deploy the reviewed artifact without regenerating it.

Approval applies to a specific code revision and input snapshot.

GitOps gives us a reviewed desired state



Reconcile only from fresh evidence and reviewed intent.

Now choose the mechanism.

The safe workflow is already defined.

CLI → structured models → YANG / OpenConfig → NETCONF / gNMI

Each mechanism must preserve the same preconditions,
service checks and recovery contract.

Netmiko: automate the CLI you already know

Netmiko manages a CLI session over SSH. An illustrative SR Linux read:

```
from netmiko import ConnectHandler

with ConnectHandler(**device) as conn:
    output = conn.send_command("show interface")
```

It handles prompts and modes.

You still have to interpret the result.

A prompt is not a postcheck.

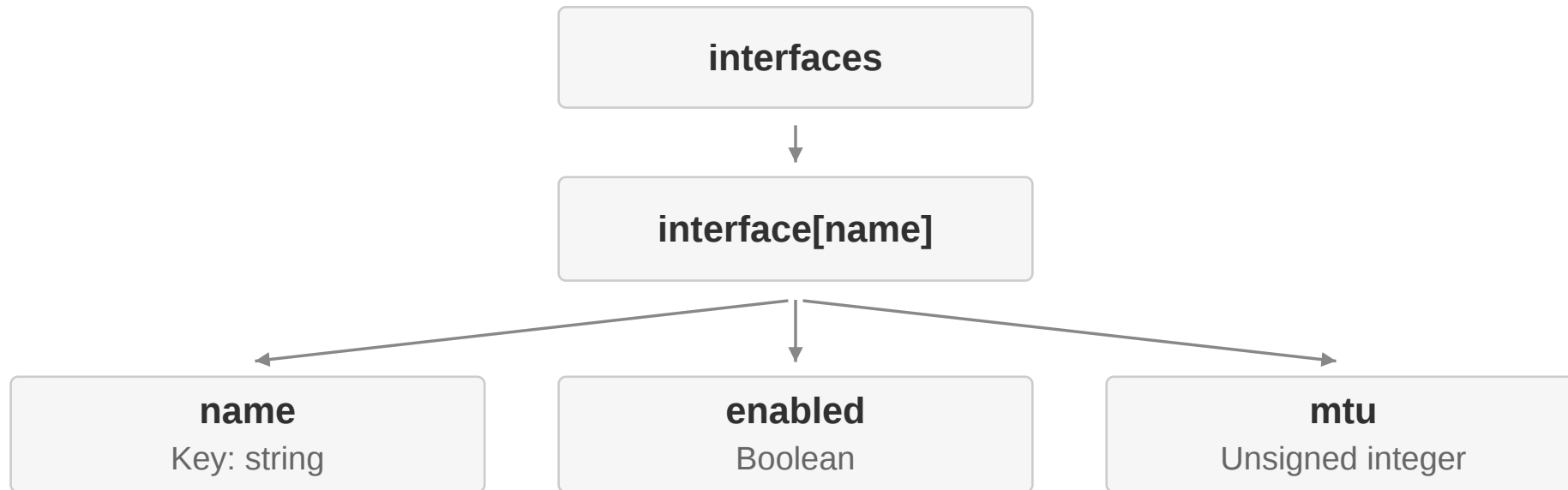
Illustrative CLI error, not SR Linux syntax:

```
router(config)# interface does-not-exist
% Invalid interface
router#
```

The session survived. The command did not.

Keep raw output. Distinguish an empty result from a parser failure.
Read state back and run the service check after a write.

YANG describes management data.



The model constrains data. The change contract constrains behavior.

YANG keys and constraints

```
list interface {  
    key "name";  
    leaf name { type string; }  
    leaf enabled { type boolean; default true; }  
    leaf mtu { type uint16 { range "1280..9216"; } }  
}
```

A module contains this fragment. The MTU range is illustrative.

A list key must name an existing child leaf.

Enabled, but no carrier

View	Interface fact
Configuration	Interface is enabled
Observed state	Carrier is down

Configuration and observed state answer different questions.

Check the physical link and the peer before changing the configuration.

OpenConfig

Vendor-neutral YANG models for interfaces, routing and telemetry.

A common model helps tools address the same concepts.
It does not guarantee identical device support.

Check the supported paths and operations.

NETCONF edits a datastore.

With a supported candidate datastore:

1. Lock, edit and validate.
2. Commit.
3. Verify the service independently.
4. Unlock and record.

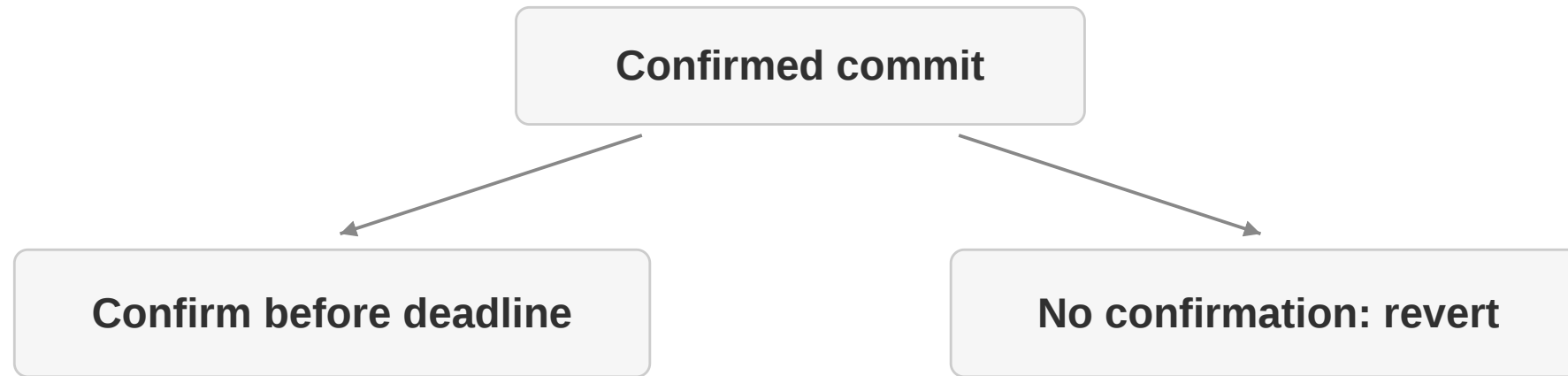
The session advertises which capabilities are available.

Read the NETCONF error.

```
<rpc-error>  
  <error-type>application</error-type>  
  <error-tag>invalid-value</error-tag>  
  <error-severity>error</error-severity>  
</rpc-error>
```

Inspect the tag, path, message and datastore state after failure.
Then decide whether a retry is safe.

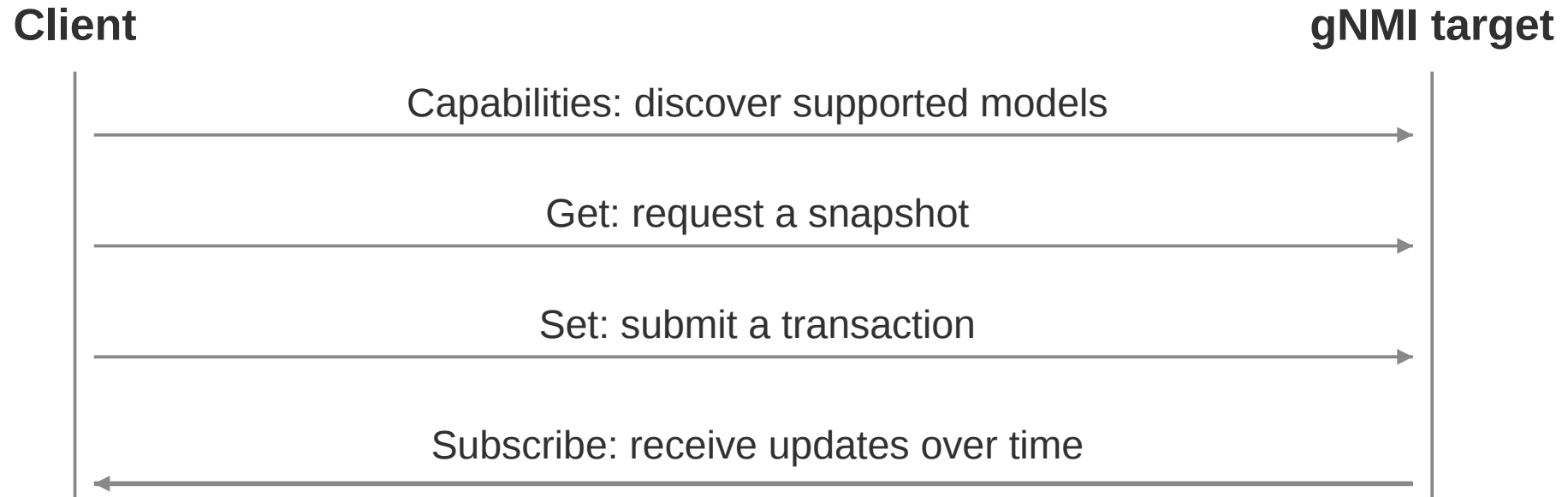
Confirmed commit



Verify service before confirming. Requires target support.

Who confirms if automation loses contact? Test the revert path.

gNMI exposes four operations.



Check which models and operations the target supports.

Read a gNMI path

```
/interface[name=ethernet-1/1]/statistics
```

- `name=ethernet-1/1` selects one interface.
- `statistics` selects its subtree.
- The **target** identifies the router.

Next: read the BOB1 policy attachment.

Read the live policy over gNMI.

From the operator workstation, read the group and instance policies:

```
BGP_PATH='/network-instance[name=default]/protocols/bgp'
POLICY_PATH="$BGP_PATH/group[group-name=lagoon-v4]/export-policy"
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSrl1!' --skip-verify \
  --encoding json_ietf get --path "$POLICY_PATH" \
  --path "$BGP_PATH/export-policy"
```

Confirm the override and inherited instance policy before writing.
Keep this shell open for the Set exercise.

gNMI Set

One request, one target, one transaction.

Result	Required behavior
Target reports success	Apply all requested changes
Target reports failure	Restore affected state to its pre-request state

A timeout leaves the outcome unknown. Read back before retrying.
Coordinate routers separately. Verify the service afterward.

Repair the same incident with gNMI.

In the same operator shell, using the reviewed `POLICY_PATH`:

```
gnmic -a edge01.bob1.reefnet.test:57400 \  
-u admin -p 'NokiaSr11!' --skip-verify \  
--encoding json_ietf set --delete "$POLICY_PATH"
```

Re-read state. Repeat HTTP from both external probes over both families.
Run `make test` in the **lab-directory terminal** for ICMP checks.

The API response is not the service postcheck.

Review a change plan

Eight minutes in groups of 3–4.

Proposed plan	
Input	Yesterday's inventory
Validation	Syntax only
Rollout	All routers at once
Success	Exit code 0
Recovery	Rerun the change

Agree on **two improvements** and their acceptance checks.

Automation: close the loop.

Two minutes. Individual notes.

A write times out after the request was sent.

- What evidence establishes the current state?
- Which service check proves the repair worked?
- What must the next operator find in the change record?

In the lab terminal, stop the runtime with `make down`.

Next: test these safeguards against real production incidents.

Who Broke the Internet?

13 October 2026. **Day 2, closing block.**

Now test the change workflow against real outages.

Everything is redundant until the shared dependency breaks.

Which safeguard would have changed the outcome?

The rules of the show

For each case:

1. who lost what, and for how long
2. the failure boundary
3. evidence that distinguishes the cause
4. the recovery dependency
5. one safeguard worth testing

We are studying mechanisms, not awarding blame.

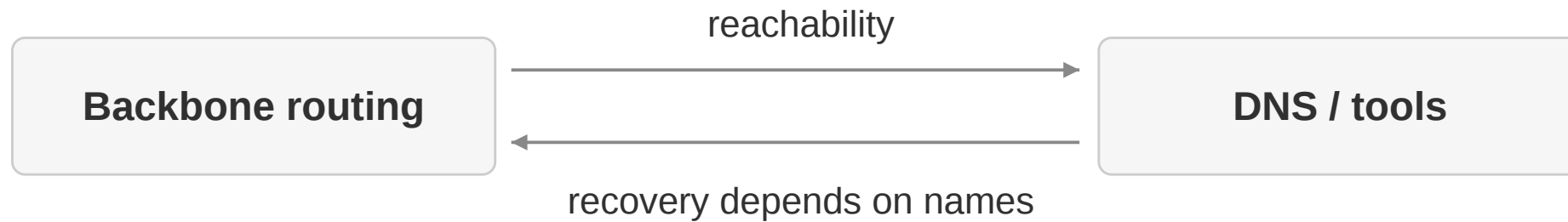
Facebook: the backbone took DNS with it

4 October 2021.

- A maintenance command was meant to assess backbone capacity.
- An audit-tool bug let it disconnect the backbone instead.
- DNS sites lost data-center reachability and withdrew their BGP routes.
- Authoritative DNS kept running, but clients could no longer reach it.

Impact: Facebook, Instagram and WhatsApp became unreachable.

Facebook: recovery lost its own tools



Break the loop before the outage.

Normal and out-of-band access failed. Engineers had to go onsite.

They restored the backbone, then brought services back under controlled load.

Can you still reach the router?

In-band: management traffic uses the production network.

Out-of-band: a separate path to the device or console.

A separate path can still share dependencies:

- power
- fiber route
- jump host
- DNS and authentication (AAA)

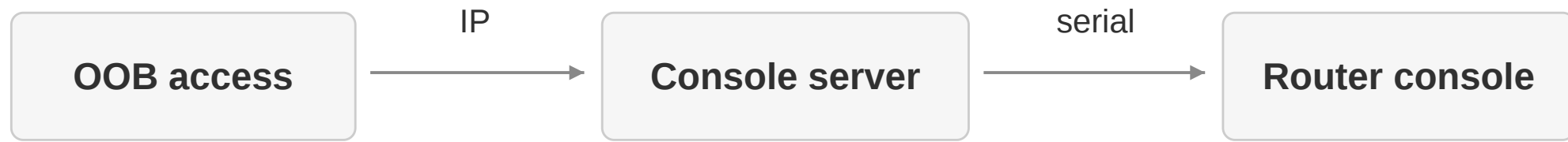
A console port is not a management port

A **console server** gives remote access to attached serial ports.

Connection on a production router	What it needs
Ethernet management port	IP connectivity and a working management service
Serial console port	correct cable, serial settings, responsive console

Serial bypasses IP forwarding. It still needs power and a responsive console.

Production recovery example: a console path



Power, emergency authentication and reachable credentials still matter.

This production recovery path is not emulated by BOB1.

Test login and repair while the production path is unavailable.

Task: the console is reachable, login fails



Exercise: AAA uses the failed backbone. Which fallback must exist?

Recovery only works if it was prepared

For the production recovery path, record each dependency.

Need	Prepared detail
Transit	circuit and provider contact
Console	server, port, cable, serial settings
Authentication	tested emergency login
DNS / secrets	reachable resolver and credentials
Power	backup for every device in the path

After login, verify that the account can actually change configuration.

Remote hands need an exact job

If remote recovery is blocked, onsite staff become your eyes and hands.

```
Location:   Frankfurt, rack R12
Equipment:  edge-a, port 2
Task:       verify label and photograph LEDs
Scope:      inspect only, no disconnect
Return:     timestamp, photos, label mismatch
```

For vendor escalation, add version, symptoms and logs.

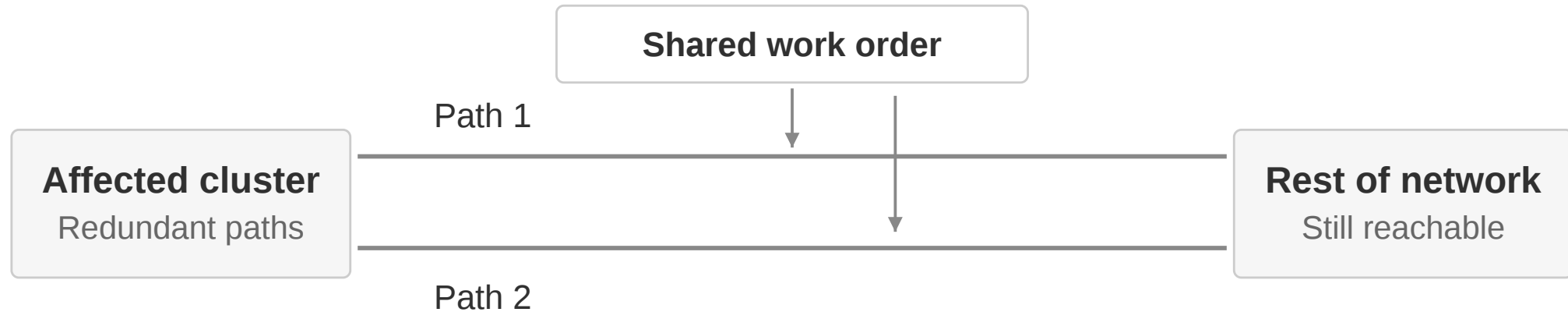
Google Cloud: incompatible optics

1 September 2026. A router capacity upgrade in **us-central1**.

- Technicians replaced optical transceivers on redundant routers.
- The new optics were incompatible with the fabric-side optics.
- The work crossed redundant routers before connectivity was checked.

Impact: affected VMs lost incoming and outgoing connectivity.
Only parts of **us-central1-b** and **us-central1-f** were isolated.

Google Cloud: one worklist defeated redundancy



1 September 2026: all affected paths were unplugged within **13 minutes**.
The worklist omitted router-by-router sequencing and verification.

Recovery: divert traffic, reinstall the original optics, verify links.

OVHcloud: undamaged did not mean available

Strasbourg, 10 March 2021. Fire began in SBG2 at 00:47 CET.

- SBG2 was destroyed. SBG1 lost four of its twelve rooms.
- SBG3 and SBG4 were undamaged, but powered down.
- Recovery involved site repairs, replacement servers and available backups.

Replacement capacity does not recreate lost data.

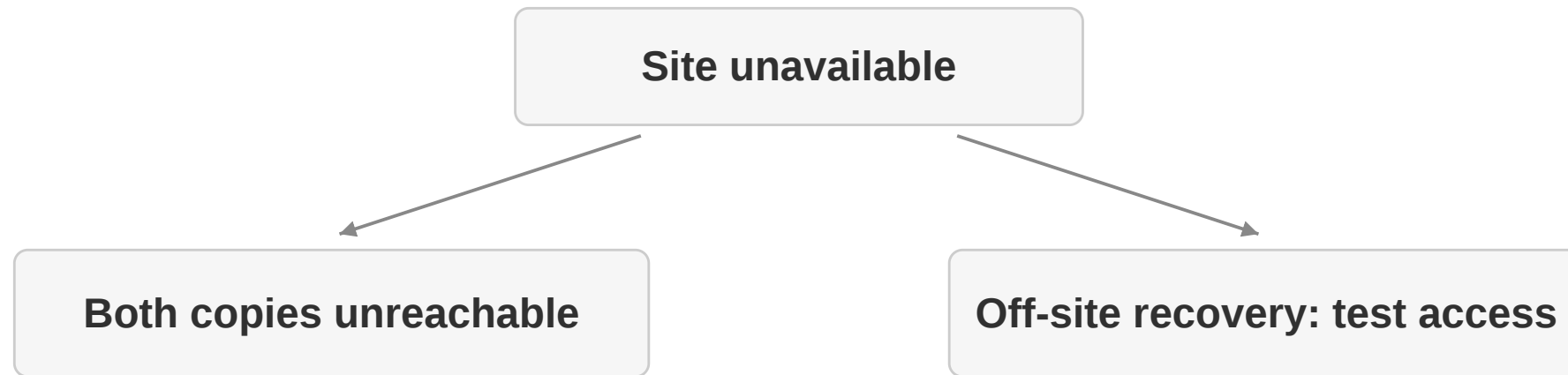
Same fire. Different recovery.

OVHcloud Strasbourg, 10 March 2021.

Service	What users lost	Recovery
Rust / Facepunch	Game data on 25 affected EU servers	Data unrecoverable
Lichess	24 hours of puzzle history	Backup from another data center

A replacement server does not bring back player progress.

Two copies, one failure domain

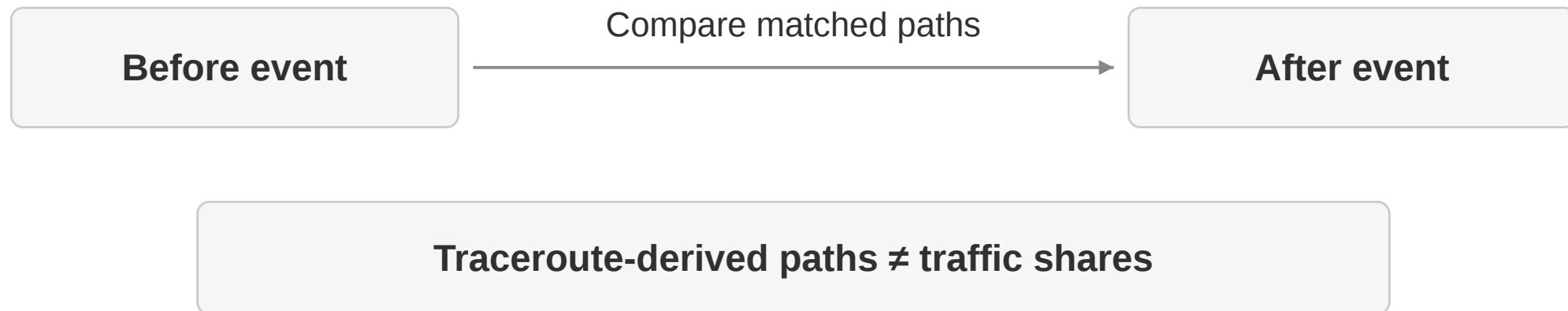


Copies do not remove a shared failure domain.

Exercise: assume both copies are inside the unavailable site.
Which recovery plan still works?

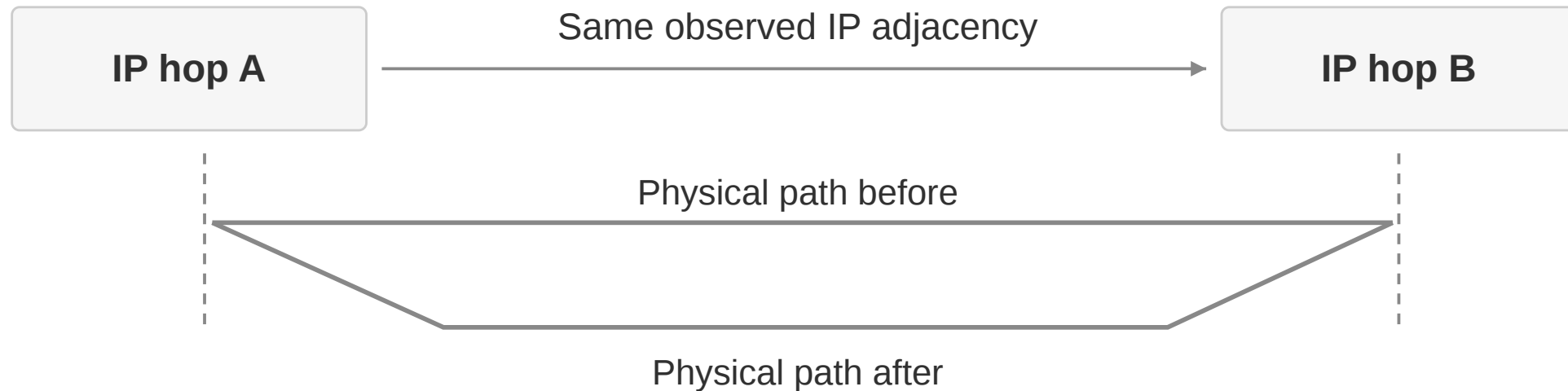
Baltic Sea: paths after the cuts

RIPE Atlas, November 2024: Germany–Finland paths.



A different physical boundary: the cable route. Compare matched observers.
This diagram shows the method, not traffic shares.

The IP path can look unchanged



Traceroute alone cannot identify a damaged cable.

Red Sea: damage is not attribution

24 February 2024: three cable systems were damaged.

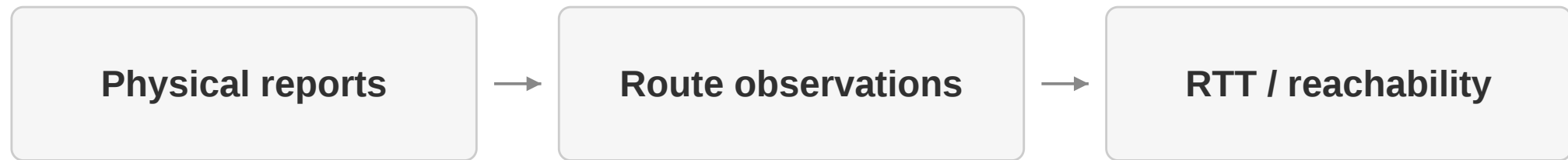
Kentik's analysis supports an anchor-dragging explanation involving the *Rubymar*.

Keep the claims separate:

- network reachability and delay: observable
- physical cause: external evidence
- deliberate attack on cables: not established by the network data

Red Sea: the cloud takes a detour.

September 2025: connectivity can continue while paths get slower.

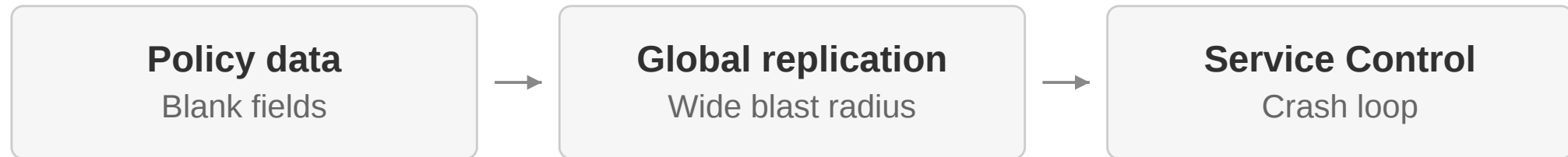


Triangulate evidence. An IP path cannot identify a damaged cable.

Which measurement separates **slower** from **unreachable**?

Google Cloud: a dormant bug goes global

12 June 2025. One policy update crosses regional boundaries.



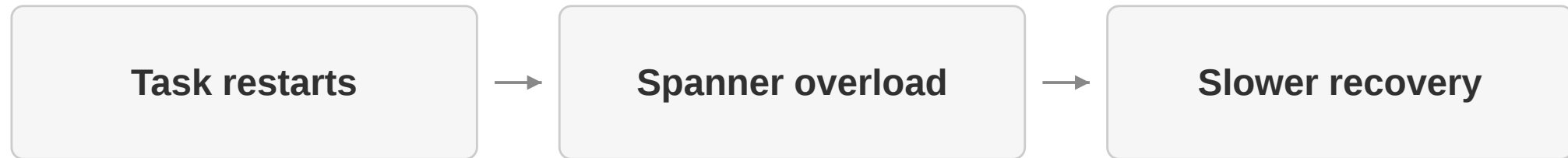
A null pointer crashed the API policy-checking process.

Service Control checks API policy. Blank fields triggered HTTP 503 errors.

Impact: API errors affected Cloud Storage, IAM and other products.

Google Cloud: recovery overloaded Spanner

12 June 2025. Disabling the faulty check stopped the crash loop.

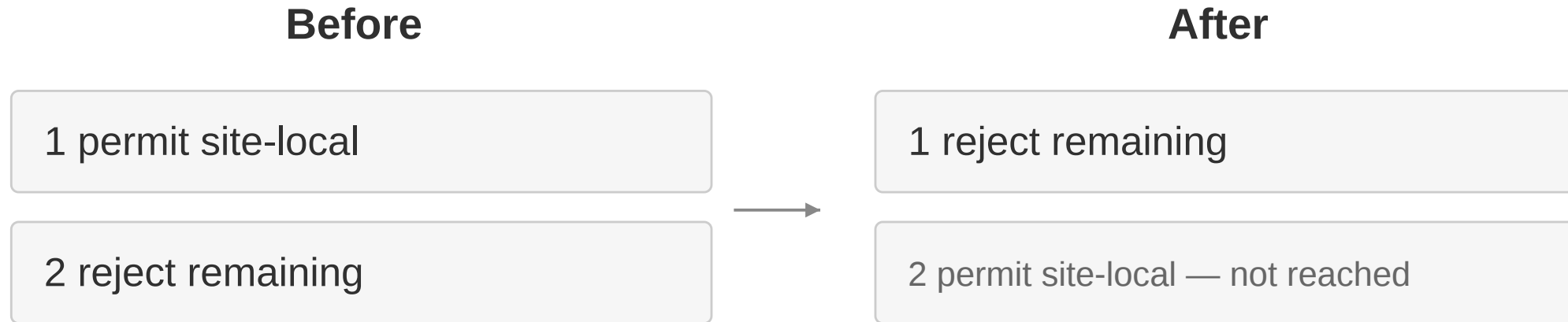


Missing randomized exponential backoff amplified the restart load.

In us-central1, Google throttled restarts and redirected database traffic.
Some customer monitoring failed too. The first status update took about an hour.

Cloudflare: policy order withdrew routes

21 June 2022. Simplified BGP export-policy order:



First matching rule wins. Later permission cannot undo rejection.

Site-local routes disappeared. Servers lost origin access.
Internal load balancing also failed.

The canary never exercised the new architecture

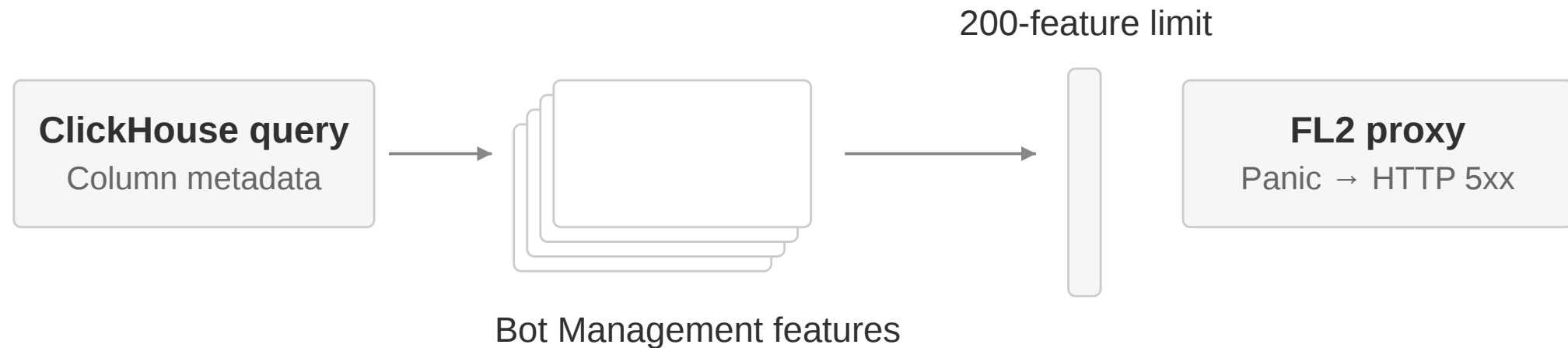
Cloudflare, 21 June 2022. Times in UTC.

- Early rollout stages used the older architecture.
- **06:27**: rollout reached 19 Multi-Colo PoP (MCP) sites.
- These sites handled about half of all requests.
- Backup access enabled rollback. The last revert finished at **07:42**.

Canary the architecture that runs the changed policy.

Cloudflare: metadata became a bad feature file

18 November 2025. A permissions change exposed more column metadata.



Without a database filter, duplicate metadata pushed the file past **200 features**. Older proxies returned zero bot scores. Blocking depended on customer rules.

Cloudflare: stop distributing the bad file

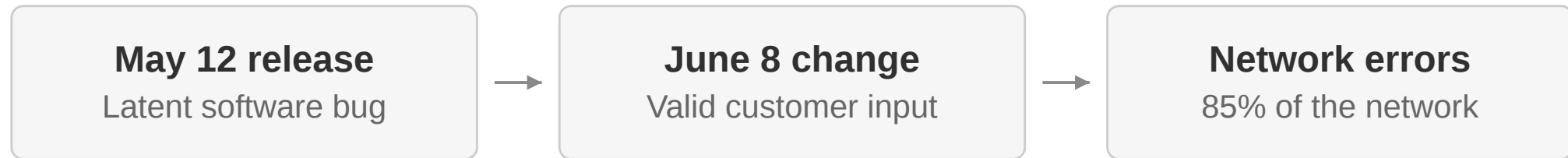
18 November 2025. Recovery milestones in UTC.

Time	What changed
14:24	Stopped generating and distributing new feature files
14:30	Known-good file restored most traffic
17:06	Remaining services recovered

Impact: websites returned 5xx errors. Turnstile failures blocked logins. Restoring the file also required restarting the core proxy.

Fastly: valid input, latent bug

8 June 2021.



The input was valid. The software was not prepared for it.

Impact: CNN, The New York Times and Reddit became unavailable.
Fastly disabled the triggering configuration. The software fix followed later.

Fastly: detection is not recovery

8 June 2021. Times in UTC.



12:35: incident mitigated. **17:25:** permanent bug-fix rollout began.
Detection, mitigation and permanent repair are different clocks.

Name the dependency.

Symptom	Pattern to investigate
Remote repair cannot start	Recovery shares the failed path
Redundant routers disappear together	Shared procedure or site
Every region fails after one update	Global data distribution
Dashboard stays green during impact	Missing or stale evidence

These are hypotheses, not diagnoses.

Your turn: redesign one boundary.

Pick one case. Draw its service path and recovery path.

Mark the shared dependency. Remove one dependency from recovery.
Propose a test that proves the revised path survives.

A second box is not a second failure domain.

From postmortem to early warning

Choose one failure pattern from today.

Which signal would have warned you?

How would you verify recovery?

14 October: monitoring and observability.

14 October 2026

Monitoring / Observability

In Operations, the customer had to tell us something was wrong.

How could the network have told us first?

What do we want to know?

Service	Network	Device
Availability	Reachability	Health
Latency	Loss	Capacity
Correct response	Routing state	Resource pressure

Choose the question before the tool.

An observation needs an expectation.

Monitoring compares observations with expected behavior.
Observability helps explain the system's state from those observations.

Question → evidence → interpretation → action

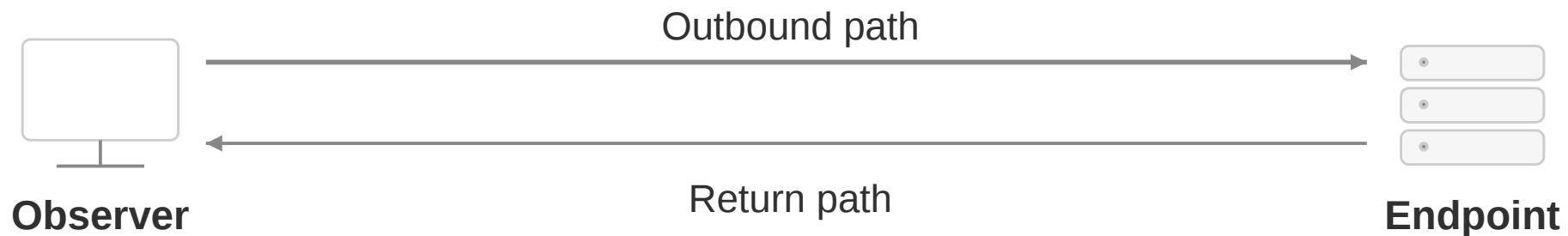
A collector transports observations. A dashboard displays them.
Neither chooses the right question for you.

Different signals answer different questions

Signal	Good for
Metric sample	how much / how fast?
Event or log	what changed?
Flow record	which traffic?
Packet capture	what was on the wire?
Active probe	can a client use the service?

SNMP and gNMI are **ways to obtain data**, not extra signal types.

Delay and loss need endpoints



RTT contains both paths and endpoint processing.

One-way delay additionally needs synchronized clocks or bounded clock error.

Choose the probe from the question

Probe	Tells us
ICMP echo	IP reachability to a responding endpoint
TCP connect	transport handshake
HTTP request	application transaction
traceroute	responding hops along a path

A successful ping does not prove the web service works.

Define the probe before you run it

One BOB1 probe is:

```
source:          probe01.bob1.lagoontransit.test
target:          198.51.100.10
address family:  IPv4
protocol:        ICMP echo
reply wait:      1 s
success:         ping exits successfully
```

The lab exporter also runs DNS probes and repeats the checks continuously.

Two checks called "reachability" can still measure different things.

The monitoring brief

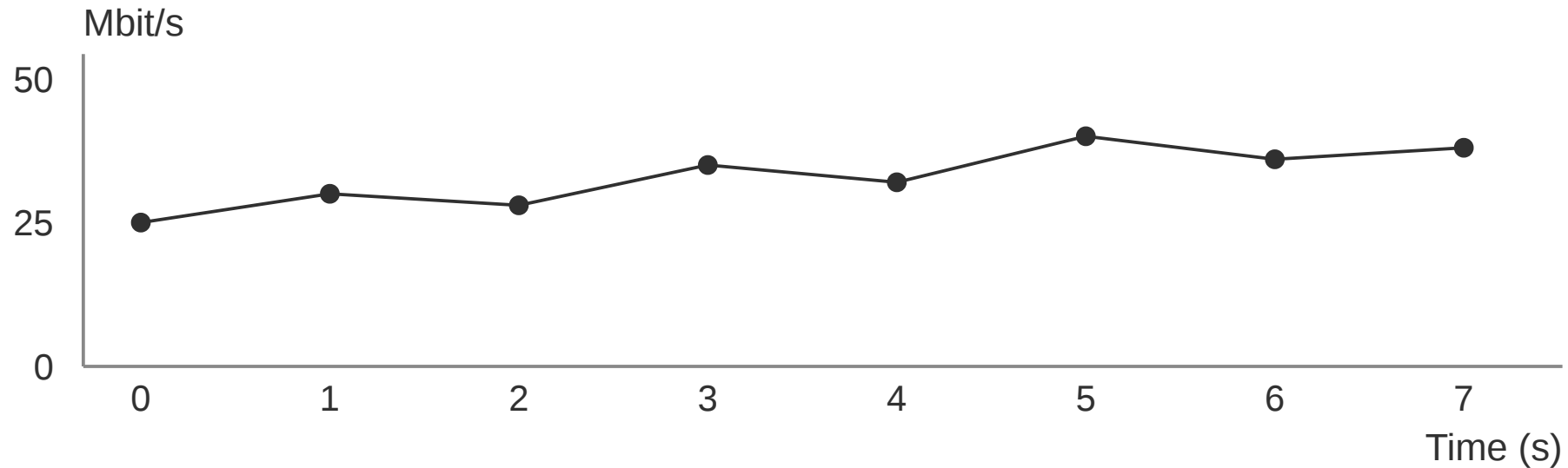
40 targets: 20 services, IPv4 and IPv6.

Two observation sources per target.

- Detect interruptions of **5 seconds or longer**.
- Stay within **20 HTTP requests/s**.
- Mark stale data **unknown within 15 seconds**.

Sizing exercise: this exceeds the small BOB1 probe set.

One ReefNet interface, one series

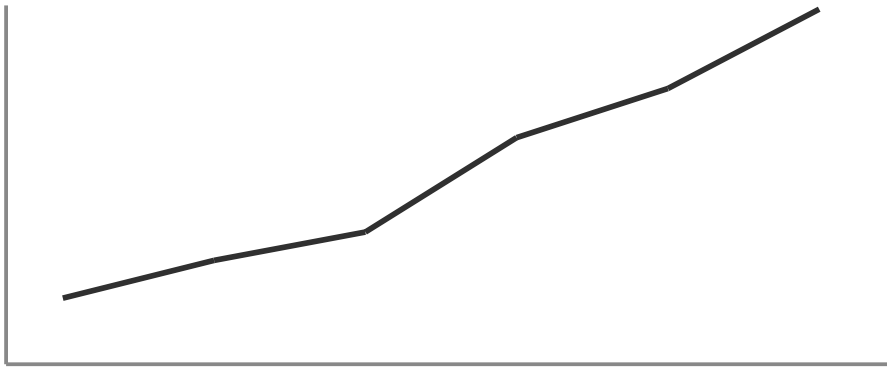


edge01, ethernet-1/3: traffic arriving from Lagoon Transit.

Each sample keeps that identity. The plotted values are illustrative.

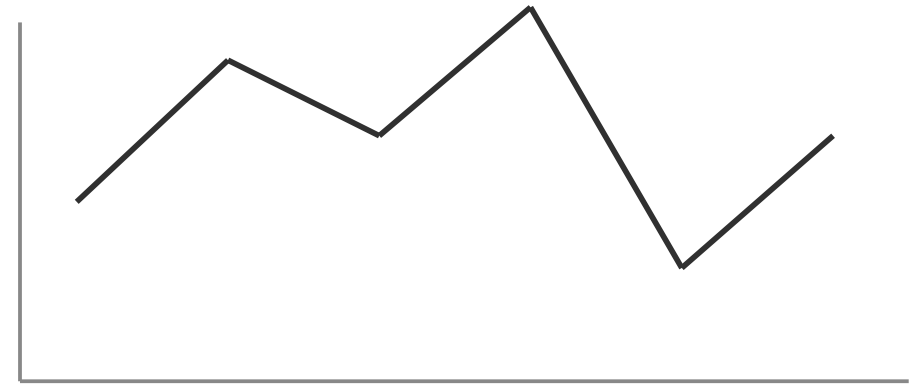
Counter or gauge?

Counter



Bytes on edge01 / ethernet-1/3

Gauge

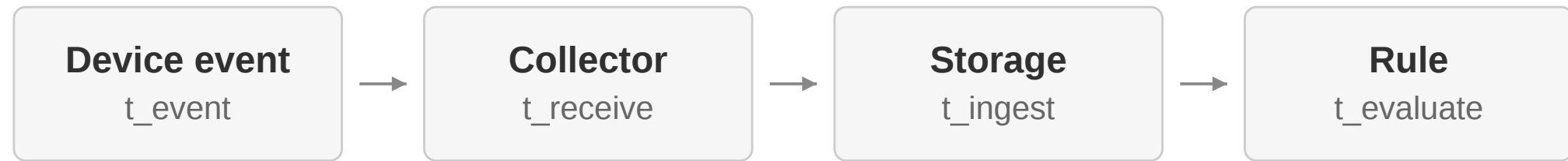


DNS-check duration from Lagoon

The byte counter accumulates. The probe duration can rise or fall.

Timestamps in the pipeline

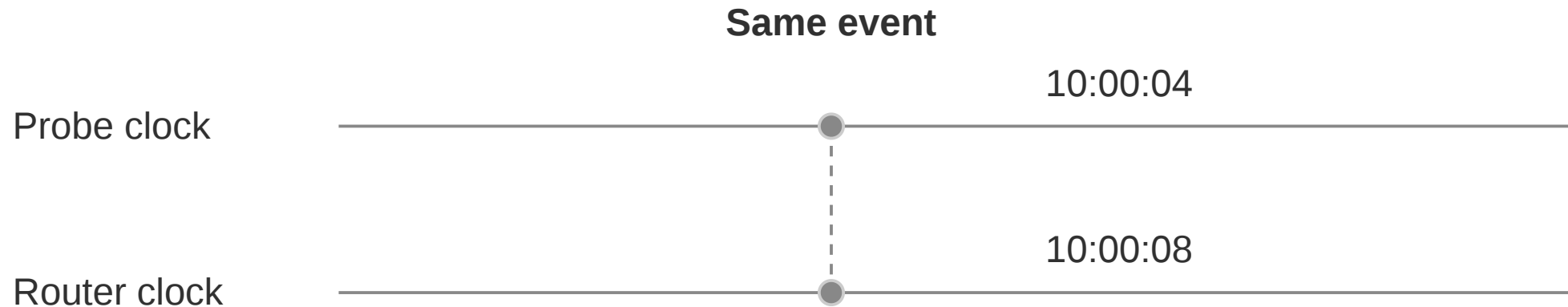
One event, several clocks.



Clock offset and transport delay are different quantities.

Subtract timestamps only after checking clock source and offset.

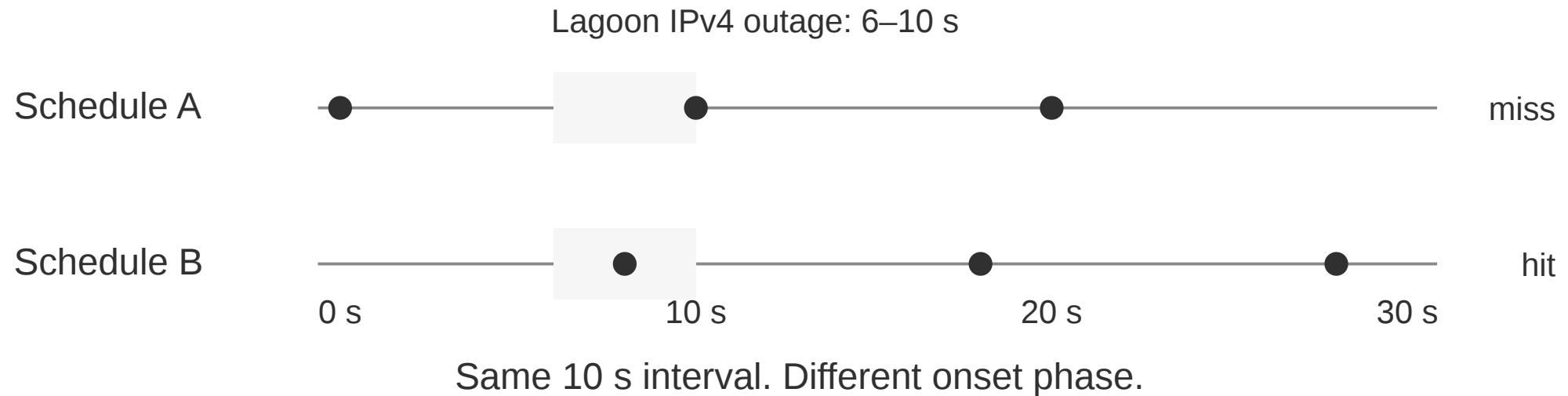
Clock error changes the story



Unsynchronized clocks distort the apparent event order.

Bound clock error before inferring event order.

Four seconds can disappear



Imagine a short Lagoon IPv4 outage. A healthy sample can miss it.

Sampling leaves gaps

An event can begin and end between observations.

- Interval and onset phase affect detection.
- Equal average rates can hide different gaps.
- A stream can carry sampled state.

Streaming periodically sampled values still leaves gaps.

Budget the ReefNet probe round

Ocean Research is checked from **Lagoon and Pacific**.
Each source tests **IPv4 and IPv6**, using **ICMP and DNS**.

$2 \times 2 \times 2 = 8$ checks per round.

Planning example: a round every 5 seconds needs **1.6 checks/s**.

In the lab, each round finishes before a 1-second pause.
What happens to the interval when a check times out?

Before subtracting counters

Check identity, timestamps and discontinuities.

A negative delta may mean **reset, wrap or bad data**.

Two values cannot reveal multiple wraps.

Invalid evidence becomes **unknown**, never a fabricated rate.

Turn ReefNet byte counters into bit/s

Use `in-octets` on `edge01`, interface `ethernet-1/3`.

$$rate = \frac{8(C_2 - C_1)}{t_2 - t_1}$$

Counter in **bytes**, time in **seconds**. Use a reset-free interval.

Compare the result with the **50 Mbit/s Lagoon handoff**.

Calculate the Lagoon ingress rate

Illustrative `in-octets` on `edge01`, interface `ethernet-1/3`.

Time	Byte counter	Reset?
10:00:00	1,000,000	Initial sample
10:00:10	32,250,000	No
10:00:20	20,000	Yes
10:00:32	37,520,000	No

Four minutes: calculate valid rates. Compare them with the 50 Mbit/s handoff.

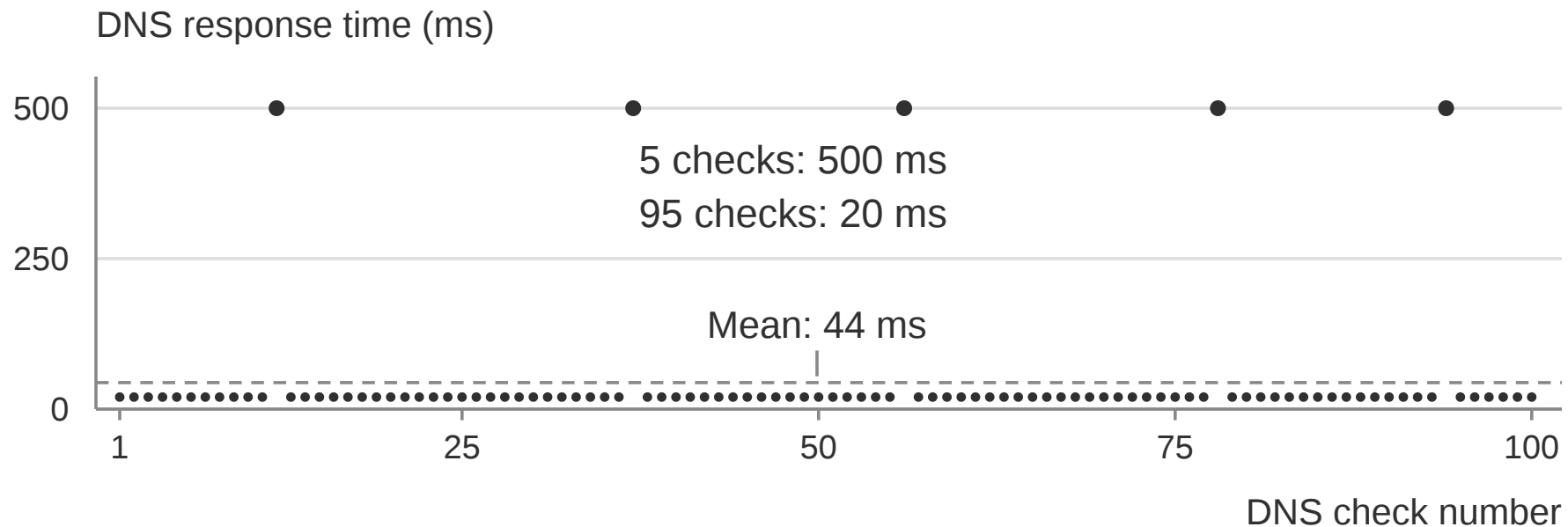
25 Mbit/s on a 50 Mbit/s handoff

Interval	Calculation	Result
00–10 s	$8 \times 31,250,000 / 10$	25 Mbit/s
10–20 s	Counter reset	Exclude
20–32 s	$8 \times 37,500,000 / 12$	25 Mbit/s

Both valid intervals use **50% of the Lagoon handoff capacity**.

The 20,000-byte reading starts a new comparison after the reset.

What Lagoon sees when it probes DNS



100 illustrative DNS checks to Ocean Research. Start with the individual waits.
Would the **44 ms mean** explain the five long waits?

Read the distribution

View	What it tells us
Median	Middle observation
p95	Threshold covering at least 95%
Histogram	Counts within ranges
ECDF	Fraction at or below each value

A percentile is not an “average slow request”.

Percentiles do not average

A separate DNS example for `data.oceanresearch.test`. Nearest-rank p95:

Source	Checks	Duration of each check
Lagoon	100	20 ms
Pacific	1	1,000 ms

Average of the two p95s: **510 ms**.

Combined observations: **p95 = 20 ms**.

Combine observations first, then calculate the percentile.

Start from a healthy network.

In the lab directory:

```
make monitoring  
make enter NODE=operations01.bob1.reefnet.test
```

This starts healthy routing, telemetry and **25 Mbit/s** background traffic.

View	Open
Grafana	<code>http://localhost:3000</code>
NetBox	<code>http://localhost:8000</code>

Grafana: **ReefNet / BOB1** folder. Refresh every 5 seconds.

SNMP: what does the device report?

Observation	OID, typed value, device, row, poll time
Limit	Polls can miss short events
ReefNet	Compare with gNMI. SNMP is not provisioned

A manager queries an agent. **OID**: object identifier.

The **MIB**, Management Information Base, defines object meaning.

Interface “up” does not prove service health.

Keep the object identity.

```
GET      request-id 41    ifOperStatus.17
RESPONSE request-id 41    noError    INTEGER up(1)
```

The OID selects an object. 17 selects an interface row.

Check response status, device identity and the row's name/alias.

After restart or replacement, verify the mapping before joining counters.

Syslog: what changed?

Observation	Time, host, severity, app and message
Limit	Missing logs do not prove health
ReefNet	SR Linux → Alloy → Loki → Grafana

Keep source and receive times. Severity is the device's judgment.

IPFIX: which traffic crossed here?

Observation	Flow keys, counts and time interval
Limit	Typical records omit payload. Sampling affects counts
ReefNet	Record exercise without a live exporter

A **template** defines the fields. Keep its exporter and observation domain.
Port 443 alone does not identify an application.

A template ID is not globally unique.

Illustrative ReefNet exporters, not provisioned in BOB1:

Exporter	Template 256 means
edge01	source, destination, bytes
edge02	source, destination, packets

Record from edge02: 203.0.113.10, 198.51.100.10, 8000

That is **8,000 packets**, not bytes.

What context must the collector retain to decode it?

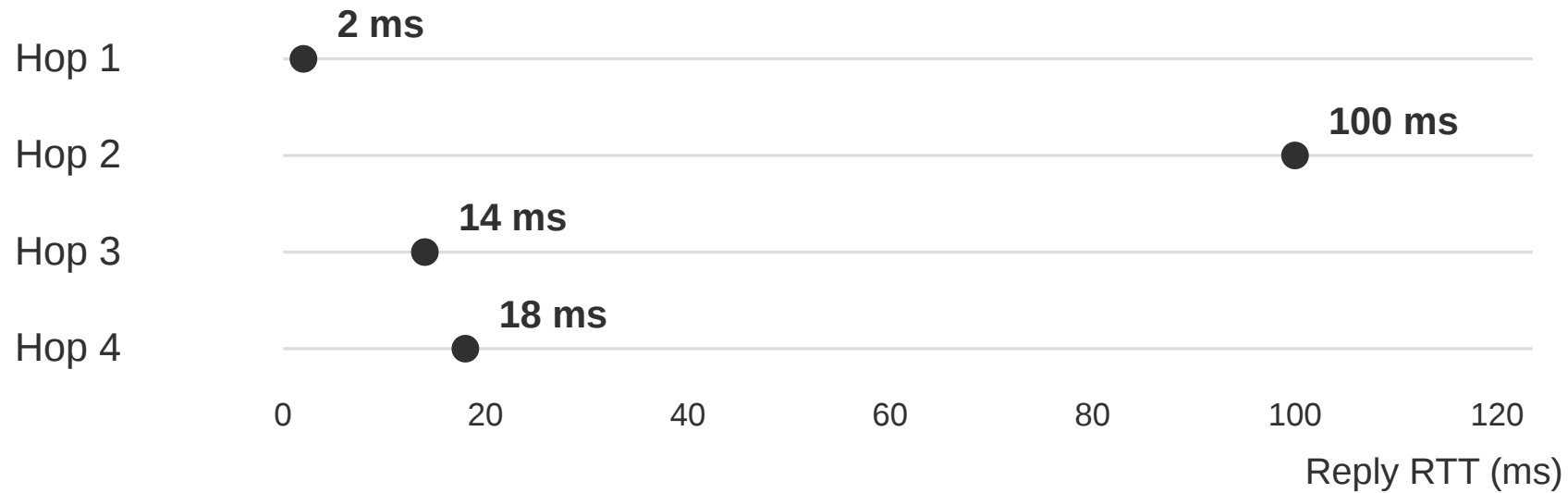
Active probes: does this exchange work?

Observation	Source, target, family, protocol, time, result
Limit	Failure does not locate the fault
ReefNet	ICMP/DNS exporters and manual HTTP checks

The DNS exporter checks command success, not the returned answer.

Traceroute: read replies carefully

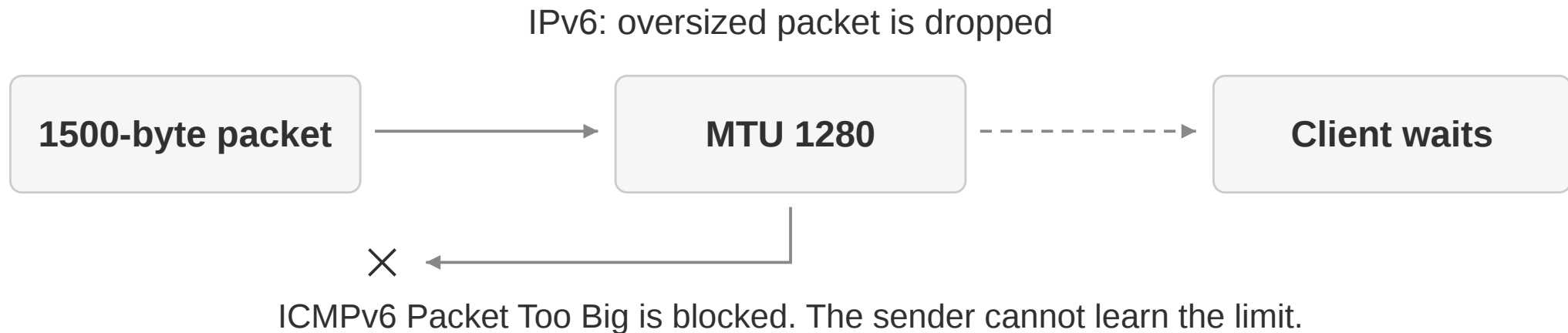
One illustrative reply per hop



Hop 2 replies slowly. It does not delay all forwarded packets by 100 ms.

Small ping, stalled download.

Thought experiment: small IPv6 responses work. Large ones stall.



Force a large IPv6 transfer after repair. Recheck IPv4.

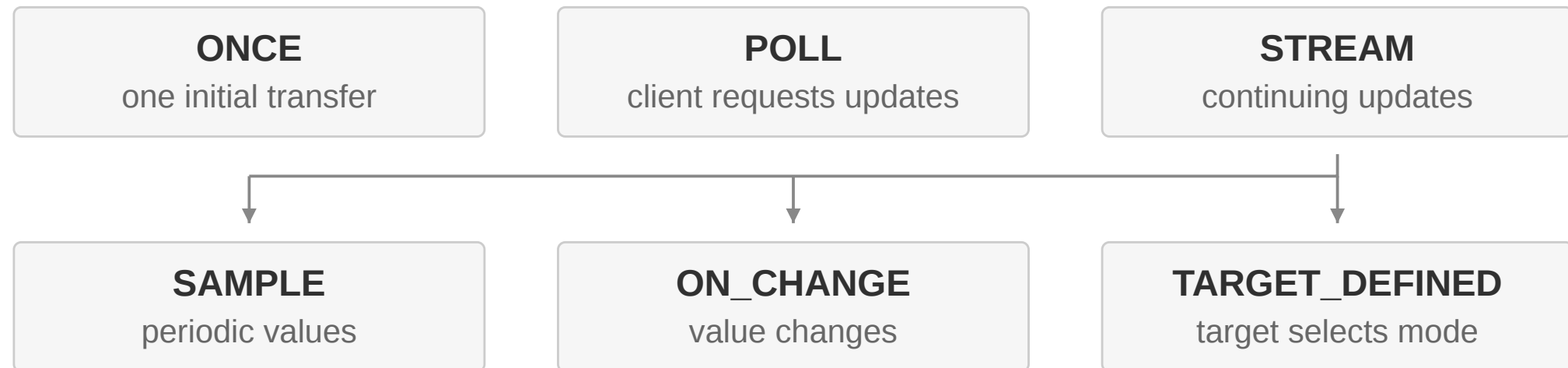
Streaming state: what changed?

Observation	Target, keyed path, timestamp, update/deletion
Limit	Reconnect cannot restore every missed event
ReefNet	gNMIc samples SR Linux state and counters

The delivery mechanism does not define freshness.

Choose the subscription semantics.

SubscriptionList mode



SAMPLE is periodic. ON_CHANGE depends on target support.

Subscribe to the live interface counters

Inside `operations01.bob1.reefnet.test`:

```
gnmic -a edge01.bob1.reefnet.test:57400 \  
-u admin -p 'NokiaSr11!' --skip-verify \  
--encoding json_ietf \  
subscribe \  
--path '/interface[name=ethernet-1/3]/statistics' \  
--stream-mode sample --sample-interval 5s
```

Watch Lagoon ingress on `ethernet-1/3`. Grafana shows its rate in bit/s.
Ctrl+C stops this manual subscription.

Change load and watch the slope

The Monitoring scenario starts with **25 Mbit/s** offered load.

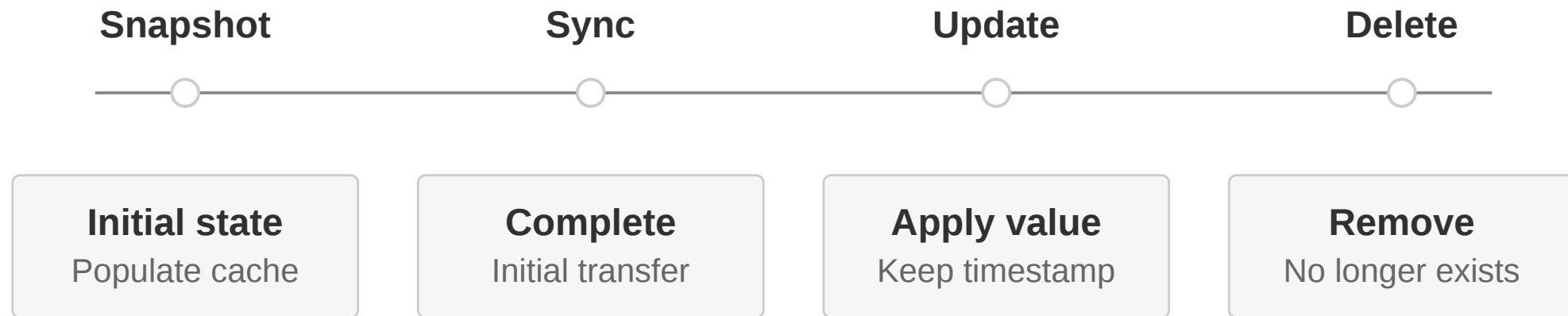
From the **lab-directory** terminal:

```
make traffic-40mbit  
make traffic-10mbit  
make traffic-25mbit
```

Run one command at a time. Observe each level for **30 seconds**.

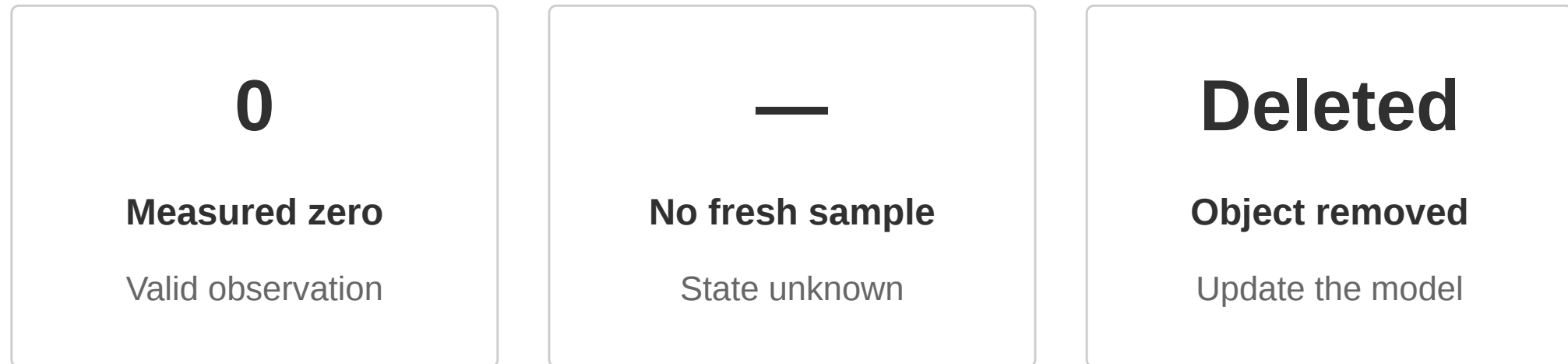
Which counter changes now? When does Grafana show the change?

Build local state from a gNMI stream



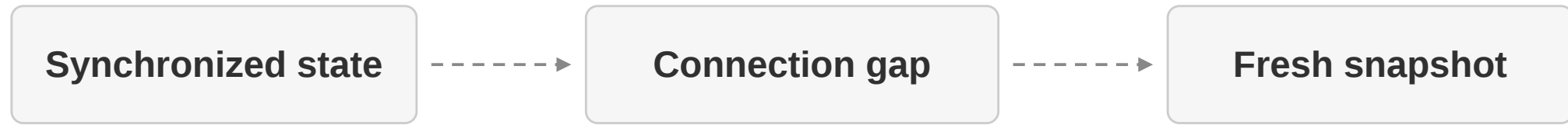
Sync completes the initial transfer. It does not guarantee future freshness.

Deletion is not zero



These states need different storage, query and alert behavior.

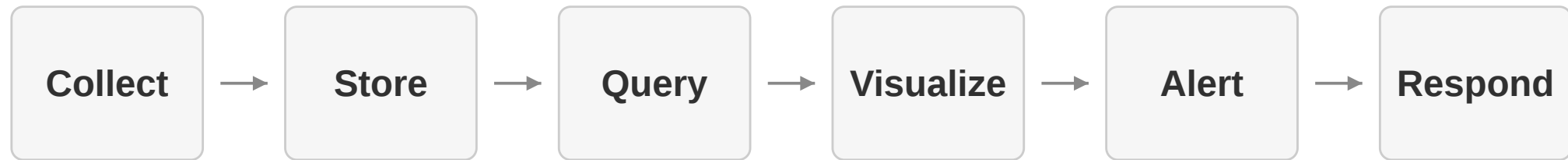
Reconnect cannot restore lost history.



Events during the gap may be irretrievable.

Rebuild current state after reconnect. Keep the observation gap visible.

Turn observations into a response.



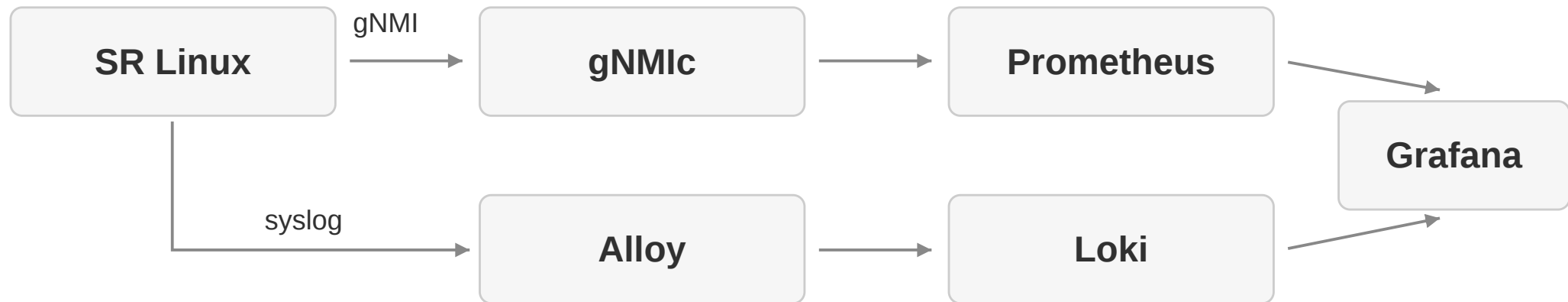
Every boundary can lose meaning, freshness or context.
Keep enough metadata to detect that loss.

Where should the collector live?

Placement	Main dependency
Central	WAN reachability
At the site	Site power and local collector
Several collectors	Coordination and duplicate handling

Test isolation. A second collector may share the first one's failure domain.

Two collection paths in ReefNet



Grafana queries metrics and logs. Each collection path can fail separately.

Prometheus stores time series.



A successful scrape can still return stale source data.

Trace a suspicious panel back to its source timestamp.

Cardinality: count ReefNet's series



8 series for one metric

One target: data.oceanresearch.test

`count(reefnet_probe_success)` should return **8** after both exporters report.

Success, duration and timestamp give **24 probe series** in total.

Would you add a request ID?

A proposed change to ReefNet's probe metric:

```
source="probe01.bob1.lagoontransit.test"  
probe="icmp"  
af="ipv4"  
target="data.oceanresearch.test"  
request_id="a-new-value-for-every-run"
```

The first four labels already identify the check.

What would the new label do to cardinality? Where should the ID go?

When would we need a queue?

Do independent consumers need **replay** from the same observations?

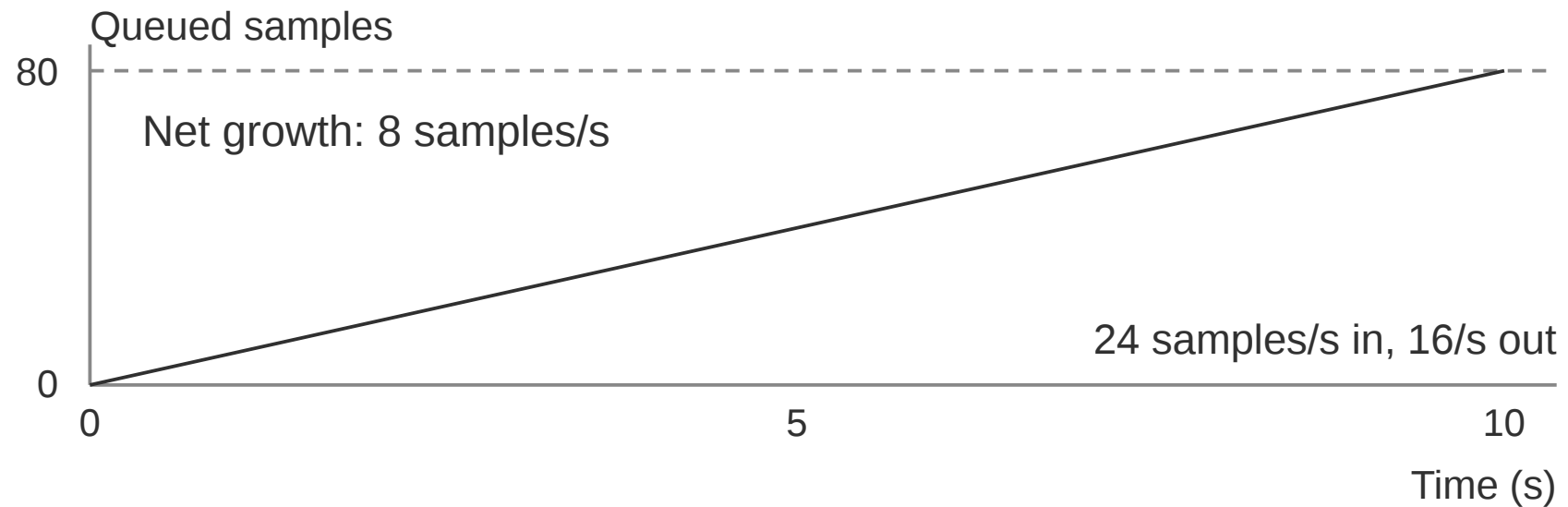
If yes, a retained event log may help.

For one dashboard and one alert path, direct storage may suffice.

Kafka also needs retention, lag monitoring and recovery.

What if ReefNet adds a slow archive?

Design exercise: forward the probe samples to a remote archive.



The buffer buys ten seconds. It cannot fix the slow writer.

Retention changes what you can ask.

Raw samples



4-s means



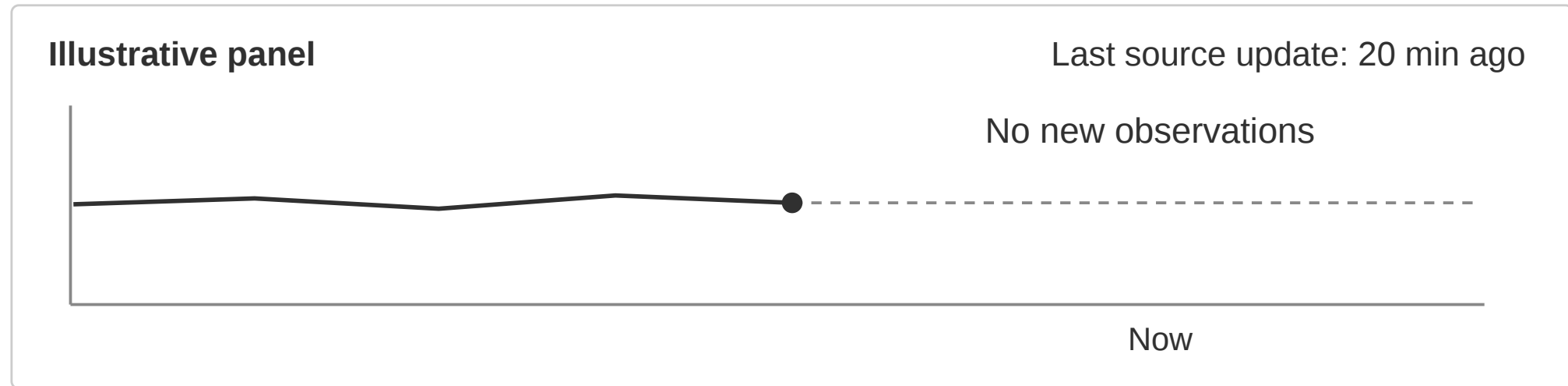
12-s mean



Same scales. Twelve seconds per row.

Aggregation saves space but can erase short failures.

Up can still mean stale.

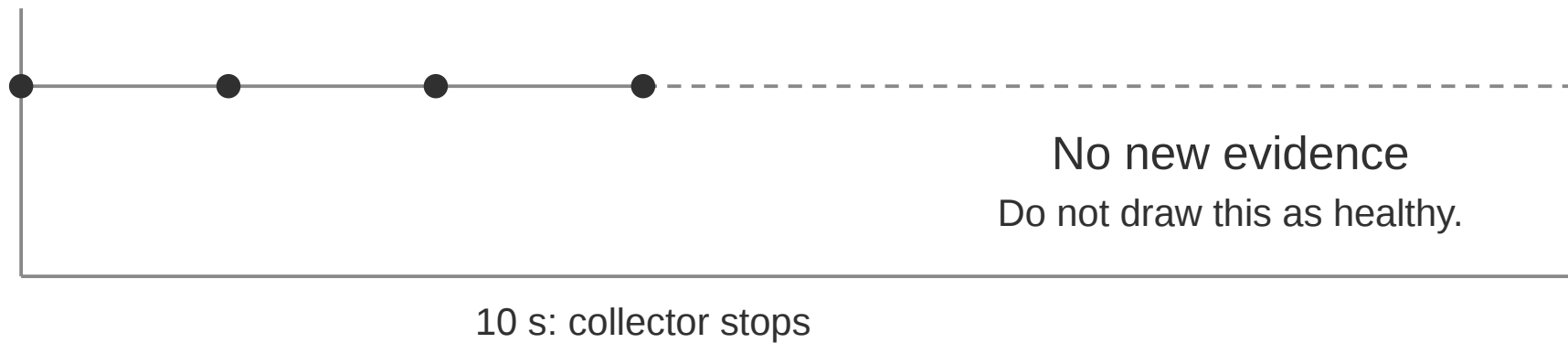


Watch source age, even when the scrape succeeds.

Healthy yesterday.

Today's status is still unknown.

Missing data on a dashboard



When should the panel stop looking healthy?

Monitor the monitoring

Component	Watch
Collector	Last valid observation
Broker / consumer	Buffer use, progress and event age
Rule engine	Errors and evaluation time
Notification path	Delivery result

Check the central monitoring service from outside.

Choose a dashboard

Question	BOB1 dashboard
Scope?	Network Overview
Link and path?	Topology & Paths
Investigate cause?	Fault Analysis
Service responding?	Service Health
Fresh data?	Telemetry Health
Router detail?	Device Detail

One question. One entity. One time window.

Same symptom, different causes

Lagoon IPv4 fails in both cases.

Evidence	Export rejected	Session down
BGP session	Established	Not established
Customer prefix exported	No	No

Session state distinguishes these hypotheses.
Next, test a different fault: an internal link goes down.

Watch a known link failure.

In Grafana **Topology & Paths**, select ReefNet core-b.
Record **admin enable + oper up**. In the lab directory:

```
make fault-link
```

On ReefNet Edge 01:

```
show interface ethernet-1/4 detail  
show network-instance default protocols ospf neighbor
```

Expect **admin disable + oper down**, then routing changes.

Admin state and oper state

The fault just injected disables `ethernet-1/4` on ReefNet Edge 01:

```
/interface[name=ethernet-1/4]/admin-state    "disable"  
/interface[name=ethernet-1/4]/oper-state     "down"
```

Admin: configured intent. **Oper:** observed behavior.

Prometheus exposes `reefnet_interface_admin_state` and `reefnet_interface_oper_state`. Read them with their timestamps.

Restore the link and watch convergence

In the lab directory:

```
make clear-link
```

Verify admin enable, oper up and the OSPF adjacency.
Correlate event time with external reachability.

The Lagoon-to-customer flow does not need `core-b`.
Its service check can remain green while the topology changes.

Capacity is not throughput

Lagoon probe

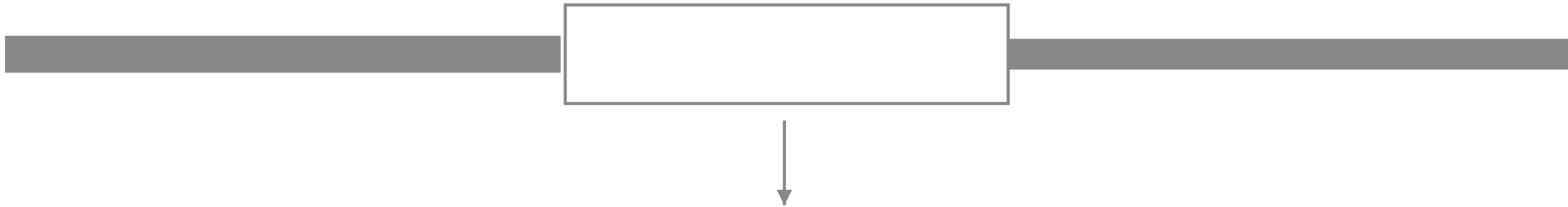
60 Mbit/s

Lagoon handoff

50 Mbit/s limit

Ocean Research

≤ 50 Mbit/s

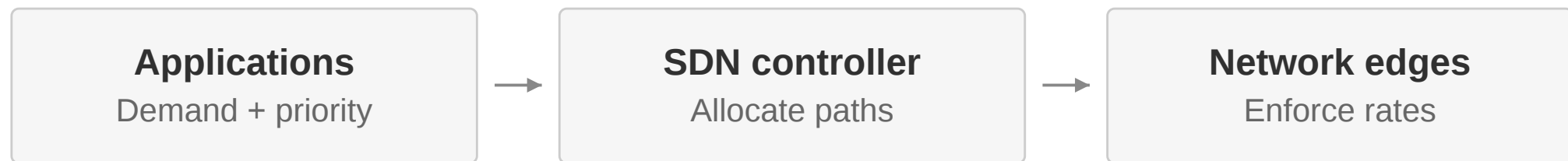


Excess: queue, drop or sender backpressure

A faster customer handoff does not remove the Lagoon bottleneck.

Google B4: busy by design

Google's private data-center WAN ran many links **near 100% utilization**.
Reported in the B4 paper, SIGCOMM 2013.

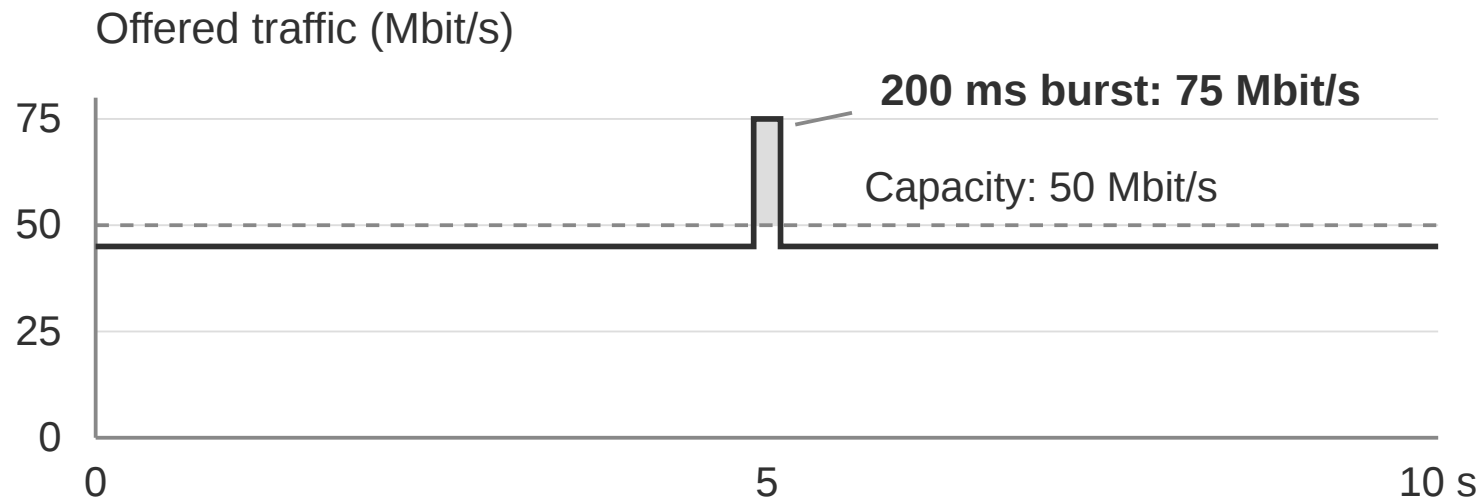


Capacity is allocated across paths as demand and failures change.

Bulk transfers yield to higher priorities. Google controls the senders, too.

High utilization needs **traffic control and a failure plan**.

90% average. Still dropping packets.



Dashboard

≈90%

mean output
over 10 s

Synthetic Lagoon example. The brief peak is lost in the average.

The burst fills the queue. **Once it is full, packets are dropped.**

Check **drop counters and queue peaks**, not just average utilization.

Cross the capacity boundary.

Transit: **50 Mbit/s**. Customer handoff: **100 Mbit/s**.

```
make traffic-40mbit  
make traffic-60mbit  
make traffic-75mbit
```

Observe each level for **30 seconds**. Compare load, rate and throughput.
Leave **75 Mbit/s** running for the next checks.

Keep route state beside the throughput evidence.

Compare offered and delivered traffic.

On ReefNet Edge 01, inspect transit and customer interfaces:

```
show interface ethernet-1/3 detail
show interface ethernet-1/1 detail
```

From the lab directory, inspect the receiving service:

```
make traffic-status
make enter NODE=service01.bob1.oceanresearch.test
tail -f /tmp/iperf-server.log
```

After recording results: **Ctrl+C**, then `exit`.

In the lab terminal, run `make traffic-10mbit`.

Keep the simulator out of the result

At 75 Mbit/s with 1,200-byte UDP datagrams:

$$75,000,000 / (1,200 \times 8) \approx 7,800 \text{ packets/s}$$

SR Linux container limit: **10,000 packets/s**.

The **50 Mbit/s transit link** should limit first. Check host load and drops.

Short bursts, long rate windows

Now isolate averaging: **40 Mbit/s stays below the 50 Mbit/s limit.**

```
make traffic-stop  
make traffic-burst-40mbit
```

This five-second burst adds bytes. A long rate window smooths its peak.

Inspect the graph. Can it recover the burst's height and duration?

Then restore `make traffic-25mbit` in the lab terminal.

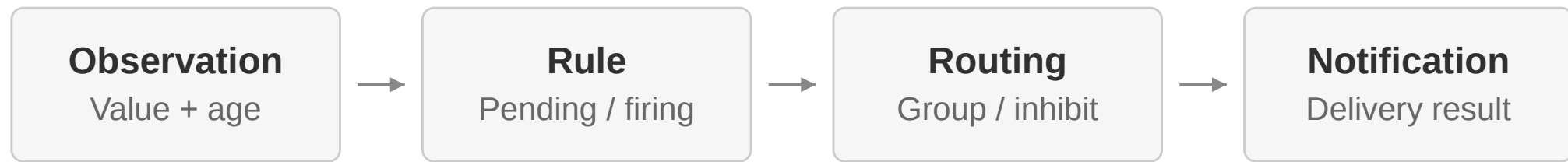
Several signals, one incident.



One controlled link fault, several observations.

Illustrative timestamps. Check clock offsets before inferring order.

The trigger is not the notification.



The rule fires, but nobody is paged. Which boundary failed?

An alert rule from the lab metric

Illustrative Prometheus rule, using the actual BOB1 metric:

```
- alert: LagoonIPv4ProbeFailed
  expr: |
    reefnet_probe_success{
      source="probe01.bob1.lagoontransit.test",
      probe="icmp", af="ipv4"
    } == 0
  for: 30s
```

The condition must stay active across evaluations for 30 seconds.
This sustained-failure rule does **not** meet our 5-second detection brief.

Alert on stale observations

```
time() - reefnet_probe_last_run_timestamp_seconds > 15
```

This detects old exported results.

Also check missing series and scrape health.

Use the expected target set to detect disappearing sources.

Flapping and notification noise

Flapping: repeated state changes.

Mechanism	Purpose
Deduplication	Skip duplicate notifications
Inhibition	Mute notifications while a related alert fires
Grouping	Combine related notifications

A firing delay reduces noise but can hide short outages.

Precision and recall

	Fault	No fault
Alert	90 true positives	99 false positives
No alert	10 false negatives	801 true negatives

1,000 illustrative observation windows.

Recall: $90/100 = 90\%$. We detect most faults.

Precision: $90/189 = 47.6\%$. More than half the alerts are false.

SLI and SLO

SLI: measured service behavior

SLO: target for that behavior over a defined window

$$SLI = \frac{\textit{successful requests}}{\textit{eligible requests}}$$

Define how missing records are handled before calculating the SLI.

An error budget for the Lagoon probe

Example target: IPv4 ICMP checks from Lagoon to Ocean Research.

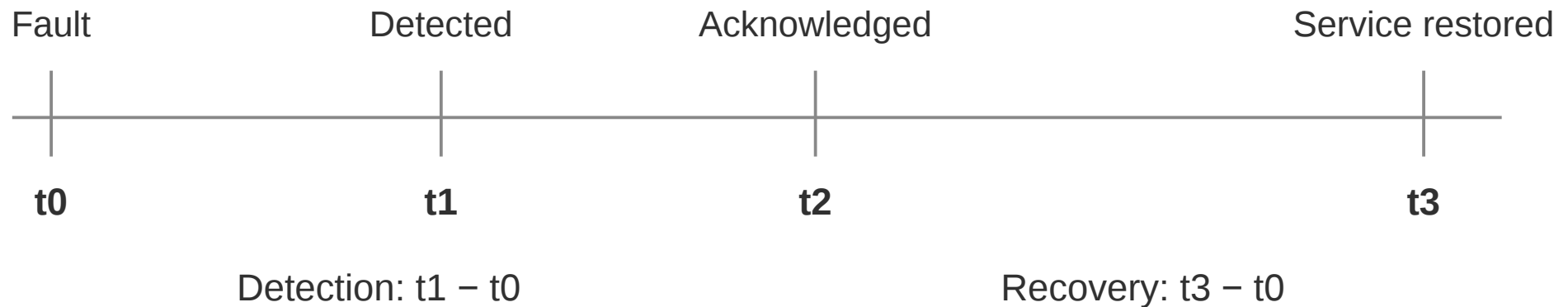
SLO: 99.9% across the next **10,000 completed checks**.

$$10000 \times (1 - 0.999) = 10 \text{ allowed failures}$$

If six checks fail, four failures remain in the budget.

Missing checks need separate reporting. This SLI describes the probe.

Detection and recovery are different clocks.



Name the endpoints before reporting a recovery duration.

Rehearse a response.

Five-minute tabletop during a change:

Vantage point	Service result
Lagoon, IPv4	Fails
Pacific, IPv4	Works

Name the first observation, incident owner and backout trigger.
Explain why that observation distinguishes the leading hypotheses.

Test the monitoring brief

Requirement	Evidence
Detect 5 s interruptions	Detection by onset phase
≤ 20 requests/s	Requests, including retries
Unknown within 15 s	Stop collection and time it
Useful diagnosis	Distinguish route loss and queue drops

Return to the **40-target brief**. Sketch the smallest design that passes.

Monitoring: close the loop.

Save the fault timeline, collection gaps and recovery evidence.

```
make traffic-stop  
make down
```

observe → question → measure → research

30 October: turn the sampling limitation into a testable research question.

30 October 2026

Research questions and sources

Monitoring gave us a limitation: short events can disappear between samples.

Now we turn that into a question we can actually test.

Leave with a testable research brief

Today we work from one observation to a study we can carry out.

1. State a question that a comparison can answer.
2. Find prior work and evidence that fit the question.
3. Choose a baseline and expose the biggest feasibility risk.

Our running example: **short outages missed between probes.**

From a problem to a research question

Observation: short failures escape our polling schedule.

Decision	Study design
Compare	Periodic and randomized schedules
Hold constant	Probe count, fault workload, endpoint
Measure	Detection fraction and delay
Bound	Tested durations and conditions

At equal cost, which schedule catches short outages better?

Write a measurable question.

Name the **factor**, **outcome** and **conditions**.

| At equal probe cost, which schedule detects more 1–10 s outages?

Define the schedules, fault distribution and detection criterion.
The comparison must allow your preferred method to lose.

What makes a scientific argument?



The claim cannot be stronger than its weakest evidence boundary.

Your contribution is the finding this evidence supports.

What kind of paper will you write?

Approach	Main contribution
Measurement study	Describe observed behavior
Comparative evaluation	Compare methods fairly
Reproduction + extension	Check and extend a published result
System or operational study	Build or investigate a concrete system

Every paper needs a question, evidence and a limited claim.

Task: challenge the claim

"Randomized polling detects outages faster."

Two minutes:

1. Name a result that would contradict this.
2. Specify outage durations, baseline and measured outcome.
3. Choose a success criterion that allows the method to lose.

Plan the measurement from the claim

Claim	Metric	Evidence
faster detection	detection delay	repeated known faults
lower load	bytes/s or requests/s	collector + target
fewer misses	detection rate	known injected events

If the measurement cannot test the claim, change the claim or the experiment.

Observation or intervention?

Observational study	Controlled experiment
polling gaps coincide with missed events	deliberately vary the polling schedule
shows association	tests an intervention

A **confounder** changes both condition and outcome.

Injecting a fault is not enough. The comparison still has to be controlled.

Find the closest comparison.

Search by **problem, mechanism and method**, then follow citations.

Record query, database, date and inclusion criteria.

Compare assumptions, baselines and evidence across the closest papers.

Your contribution changes one justified assumption or tests a new boundary.

“Nobody has done this before” is not a literature review.

Read in three passes.

Pass	Record
Relevance	Question, main result, contribution
Method	Signal, fault model, baseline, experimental unit
Challenge	Missing evidence, alternatives, transfer limits

Write one accepted claim, one limitation and one next question.
A blank comparison cell is a reading task, not automatically a gap.

Read the claim before the graph.

Trinocular studies Internet reliability through adaptive probing.
Its unit is an analyzable **IPv4 /24 block**, not every application.

Read for	Question
Scope	Which blocks and vantage points are included?
Method	Outage definition and probe budget
Validation	What independent evidence checks the detector?

A reachable block does not prove every host or application works.

Broader networking research

Venue	Typical contribution
<u>ACM SIGCOMM</u>	networking mechanisms, architectures and systems
<u>USENIX NSDI</u>	implemented networked systems
<u>IEEE INFOCOM</u>	protocols, algorithms and performance
<u>ACM CoNEXT</u>	experimental networking systems
<u>IFIP Networking</u>	protocols, systems and network performance
<u>IEEE GLOBECOM</u> / <u>ICC</u>	broad communications and networking

The venue follows the **contribution**, not just the topic.

Who develops Internet technology?

IETF

Working Groups engineer interoperable Internet protocols and standards.

IRTF

Research Groups study long-term questions about Internet technology.

Where do operators compare notes?

RIPE Meetings

Routing, addressing, measurement and Internet coordination

DENOG and NANOG

Backbone operations, routing, peering and incidents

NAF and AutoCon

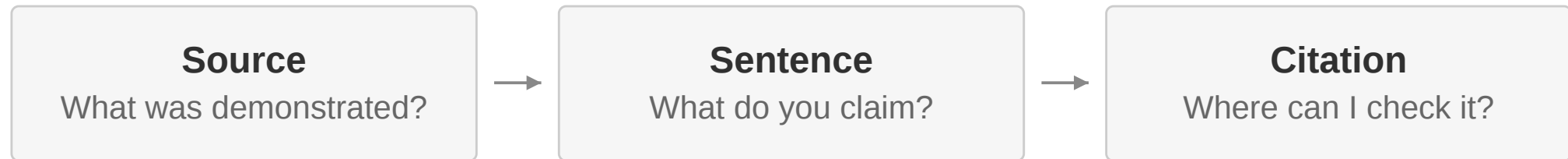
Network automation, orchestration and operational tooling

Match the source to the claim.

Claim	Evidence source
Required protocol behavior	Specification and exact section
Implemented feature	Versioned code or official documentation
Measured improvement	Paper and evaluation data
Production failure	Attributed operator report

For RFCs: check status, updates, errata and normative language.

Does the citation support the sentence?



Match the sentence to the evidence in the source, not merely to its topic.

Paraphrase the idea, not the word order

Approach	Problem or purpose
Replace a few words in the source	Still tracks the original wording
Read, take notes, explain in your argument	Paraphrase with attribution
Preserve exact wording in quotation marks	Direct quotation, with location

Use quotation marks for copied wording, even with a citation.

A DOI is not evidence.

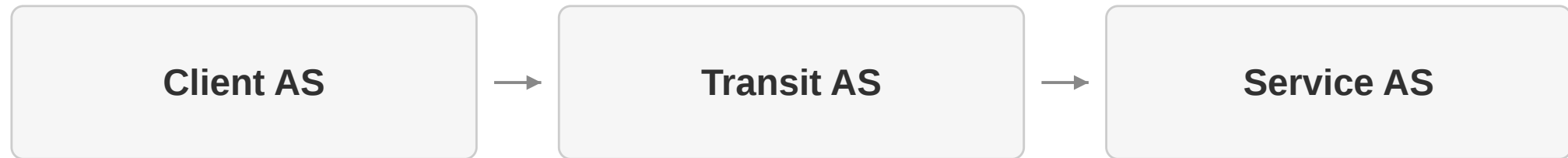
Read the result. Then check the claim.

Make evidence traceable.

Artifact	Record
Software	Version or commit and repository
Dataset	Release, identifier, access conditions
Documentation	Exact page, version and section
Figure	Data and units, or source and adaptation

Verify imported reference metadata. Mark synthetic examples.
Citation does not replace reuse permission.

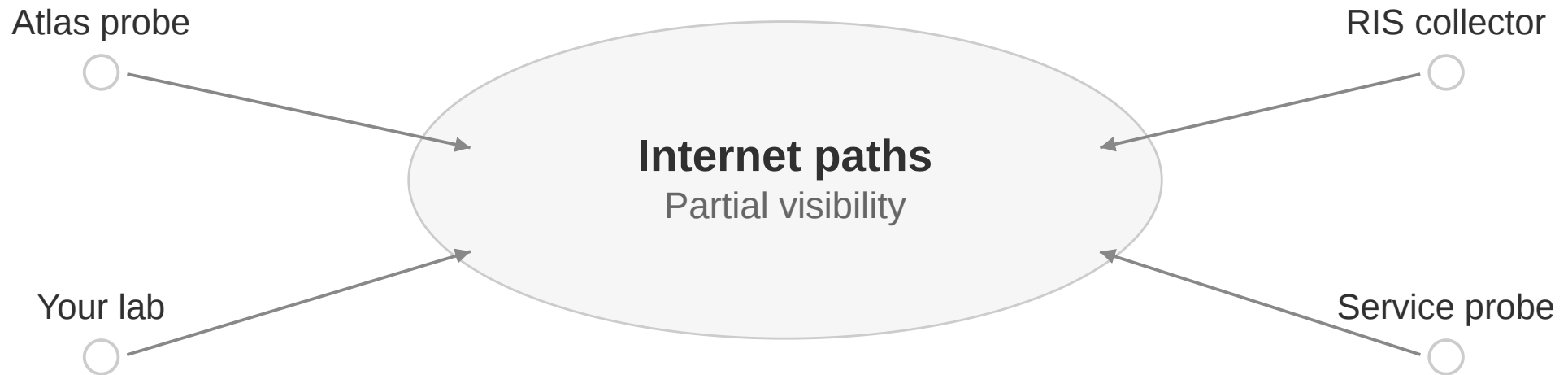
Can public data answer your question?



Each observer sees only part of this path.

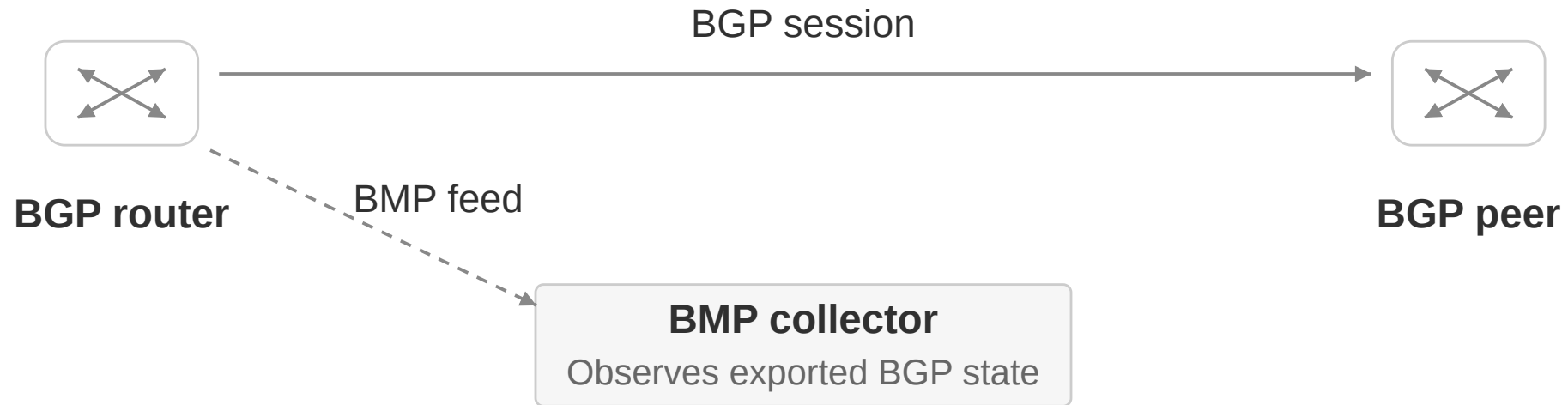
If your question extends beyond ReefNet, consider public observations.
Their coverage and resolution must fit the claim.

Public data gives us another vantage point



RIPE Atlas: active probes. **RIS:** public BGP observations.
Neither covers every Internet path.

BMP observes BGP state.



With router access, the **BGP Monitoring Protocol (BMP)** exports BGP state. It is a collection protocol, not the public RIS dataset.

Read one public routing record.

Which routes were visible for AS8966? Query public RIS history:

```
curl -sS -G https://stat.ripe.net/data/routing-history/data.json \
  --data-urlencode resource=AS8966 \
  --data-urlencode starttime=2025-09-05T00:00:00 \
  --data-urlencode endtime=2025-09-07T00:00:00 \
  --data-urlencode include_first_hop=true \
  --data-urlencode min_peers=10
```

Inspect prefix intervals, origin/first-hop pairs and RIS peer counts.
Route visibility is not traffic volume.

What did the collector actually see?

Selected fields recorded in the original course material (18 September 2026):

```
origin + first hop: 1273 8966
prefix:            2.16.44.0/24
interval, UTC:     5 September 2025, 00:00-07:59:59
full_peers_seeing: 64
```

The pair includes a first hop. It does **not** mean two origin ASes.

The query omits routes seen by fewer than ten full-feed peers.

Does route visibility prove service availability? Does 64 mean 64 users?

Make the public measurement reproducible

Keep the details another person needs to repeat it:

- UTC time window
- unaffected comparison
- query or measurement IDs
- filters and thresholds

Save them with the conclusion.

Plan responsible measurement.

Prefer existing public data or your own lab target.

Before active probing, agree on target, authorization, rate, duration and stop condition. Follow the platform's rules.

For collected data, define fields, access, retention and deletion. Network records can expose communication patterns.

Collect only what the research question needs.

Build an argument, then audit it.

Organize related work by the question the studies answer.

For each factual sentence, identify its role:

- our result → method and evidence
- prior result → original source
- protocol behavior → specification
- interpretation → inference and limitation

If you cannot classify it, check whether it belongs.

Project checkpoints

Date	Bring
30 October 2026	Question and initial source
13 November 2026	Baseline, data access, open questions
27 November 2026	Results or a concrete blocker
11 December 2026	Comparison, outline, January plan
22 January 2027	Read, write and hand in the paper review
5 February 2027	Presentation and reproducible evidence

Work in progress is welcome. **No extra graded reports.**

Your question, your next step

Write your **Research Brief**: question, closest prior work, baseline, available evidence and biggest feasibility risk.

Use today's project time to check that you can access the evidence.

13 November: turn the question into a controlled experiment.
Bring the baseline, data access and open questions.

13 November 2026

Experimental design

Now the sampling question becomes a controlled comparison.

Assign treatments to runs.

Treatment: the sampling policy.

Outcome: detection and delay.

Experimental unit: one reset run assigned a treatment.

Keep fault workload, probe budget and reset conditions equal.

Ten samples from one run are not ten independent experiments.

Fix the comparison before running it

600-second runs with **60 probes per method**.

Method	One probe in each 10 s slot
Periodic	At the start
Randomized	Uniform random times with a recorded seed

Use the same fault schedule for each pair.

Compare detection fraction and delay among detections.

Specify the experiment boundary.

Use the smallest setting that can test the claim.

Record nodes and versions, link capacity and delay, addresses and policy, probe locations and control path.

Keep control reachable when the workload fails.

Document any dependencies shared by control and experiment.

Two identical diagrams can conceal different experiments.

Virtual experiments have limits.

A virtual network can demonstrate mechanisms under controlled conditions.

It does not automatically reproduce:

- hardware queueing and forwarding-chip timing
- optical behavior
- production-scale convergence
- carrier failure domains
- Internet-wide path diversity

State which conclusion depends on virtualization.

Measure an independent reference.

The detector under test cannot define the ground truth.

Evidence	What it tells us
Experiment controller	When the fault was requested
Independent reference probe	When traffic was actually affected
Evaluated detector	Whether and when it reported the fault

A 100 ms reference interval limits onset-time resolution.

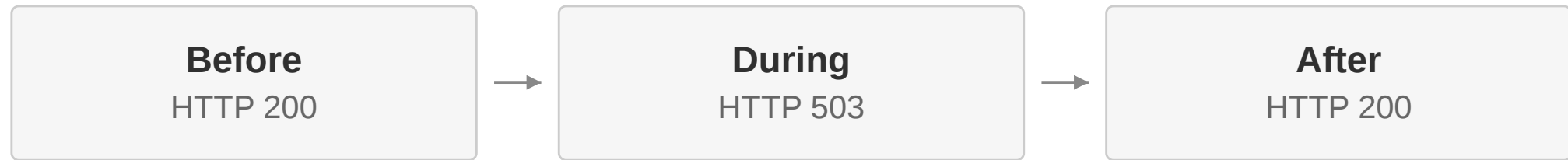
Predict the observations.

ReefNet rejects customer IPv4 on export toward Lagoon Transit.
Links, BGP sessions and management access remain up.

Predict	Compare
Customer advertisement	Lagoon vs Pacific
External reachability	Both sources, both address families
Interface rate and discards	Routing fault vs congestion

Which observation would contradict your diagnosis?

Control experiment: the endpoint fails



Same routes and interfaces. Different endpoint behavior.

An HTTP 503 response is not evidence of a routing failure.

Pair conditions, randomize order.

Replay each fault schedule for both methods at equal probe cost.
Reset between runs. Randomize which method runs first.

For jointly detected events: **difference = delay A – delay B.**
Report misses separately.

Block by host or workload and record contention.
Pairing does not remove clock error or unequal host load.

Startup or steady state?

Research question	Treatment of startup
How long until service recovers?	Startup is part of the outcome
What is steady-state overhead?	Apply a stated warm-up rule
Does cache warming help?	Compare defined cache conditions

Cold caches and convergence are real. Excluding them changes the experiment.

Decide how a run can fail

Outcome	Stored result
Fault detected	Detection flag and delay
Fault missed	Detection flag false, no invented delay
Probe process failed	Instrumentation failure
Reset did not restore state	Invalid run with a recorded reason

Choose these categories before looking at which method wins.

Turn the lab into an experiment



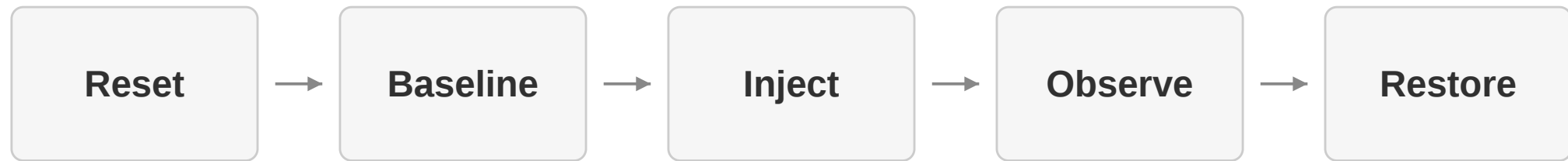
Pacific and IPv6 remain control observations.

Record the requested fault time and the first failed service check.
Keep measuring until the service is restored.

Config-change time is not traffic-failure time.

Define one experimental run

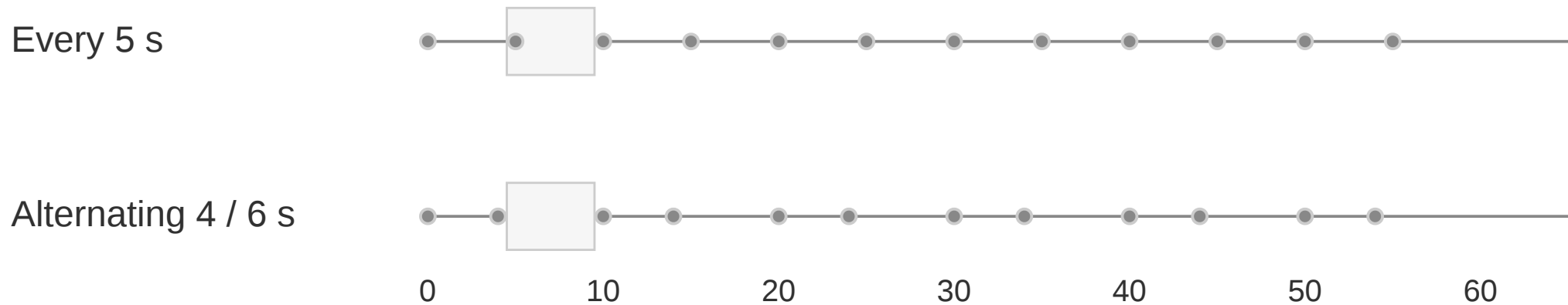
For the sampling comparison, repeat the same controlled sequence.



Run ID, seed, versions and raw results travel with the experiment.

A failed reset invalidates the comparison. Record it as a failed run.

Same budget. Different gaps.



Twelve probes per minute. A five-second fault can fit between probes.

Timing model, not the 10 s study: **5 s mean spacing**, 120 fault phases.

Compare detection and delay

Synthetic timing example: 120 onset phases, five-second faults.

Probe spacing	Detected	Missed	Median delay when detected
Every 5 s	120	0	2.50 s
Alternating 4 s, 6 s	108	12	2.25 s

The lower median ignores the missed faults.

A real experiment must also account for request duration and scheduling.

Task: compare detection delays

Separate arithmetic example: five synthetic pairs, delays in milliseconds.

Method	Pair 1	Pair 2	Pair 3	Pair 4	Pair 5
A	11	12	10	13	54
B	17	18	16	17	18

Four minutes: Which method is usually faster? Which is more predictable?

Choose a summary. State what five pairs cannot establish.

One outlier changes the mean

Statistic	A	B
Mean	20.0 ms	17.2 ms
Median	12 ms	17 ms
Range	10-54 ms	16-18 ms

A is earlier in 4/5 pairs.

Its fifth run reverses the mean.

Small sample. This only illustrates the calculation.

Five pairs. Plenty of confidence.

Exact arithmetic. Still a small sample.

More samples do not fix bias.

Source	What to examine
Sampling uncertainty	Variation between runs
Measurement uncertainty	Clocks and instrument resolution
Bias	Unequal workload or target conditions
Instrumentation effects	Polling, capture and logging overhead

More runs can reduce sampling uncertainty.
They do not fix a bad clock or an unfair comparison.

Four types of validity

Type	Example problem
Construct	DNS success is treated as HTTP service health
Internal	only one method runs on an overloaded host
External	one virtual topology stands for all networks
Statistical conclusion	correlated samples are treated as independent

Reproducibility does not remove bias.

Task: review the experiment

Method	Host	Time
A	Idle	Morning
B	Shared and busy	Afternoon

Four minutes:

- which effects are mixed together?
- what would you hold constant?
- what would you randomize?
- what would you record?

27 November 2026

Analysis and scientific argument

From paired runs to a defensible comparison.

The graph is not the result. Check the evidence behind it.

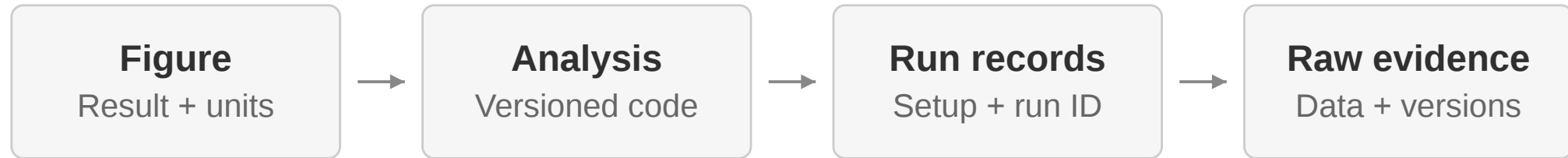
Keep the mechanism visible.

raw runs → detection counts → delays → uncertainty → claim

Before aggregation, check matched conditions and individual traces.
An average can hide short faults, load changes and a dominant run.

Keep missed events in the detection denominator.
A detector does not get faster by deleting its misses.

Follow one result back to its inputs



Follow one reported number all the way back.

Choose a number from your draft. Can another team reproduce it?

Keep one row per run

Run	Condition	Method	Detected	Delay, ms
17a	17	Periodic	Yes	820
17b	17	Randomized	No	—
18a	18	Periodic	Yes	240
18b	18	Randomized	Yes	510

Illustrative study log: one outage and 60 probes per run. Pair by condition ID.
A miss stays in the detection denominator.

Missing, missed and timed out are different

Outcome	Meaning
timeout after 2 s	no completion observed by deadline
probe never ran	service outcome unknown
event not detected	miss for this detector
run excluded	rule + reason recorded

Dropping timeouts can make a broken service look fast.

A fast median can hide missed faults

First compare coverage. This separate example has **20 faults**.

Method A: 18 of 20 detected

Median among detections: 1.0 s

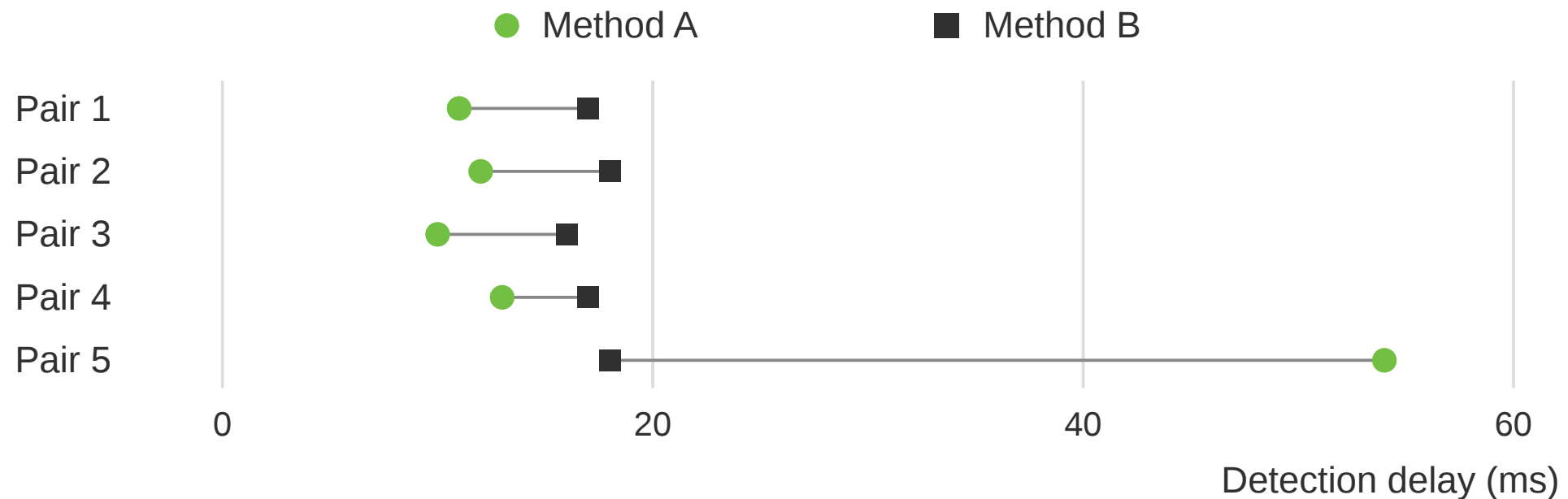
Method B: 4 of 20 detected

Median among detections: 0.5 s

Synthetic example

Compare the same detected events before claiming that B is faster.

Compare within each pair.



Return to the five synthetic pairs from **13 November**, methods A and B.

Work with paired differences.

Pair	A, ms	B, ms	A – B, ms
1	11	17	–6
2	12	18	–6
3	10	16	–6
4	13	17	–4
5	54	18	+36

Mean: **+2.8 ms**. Median: **–6 ms**. A is earlier in four pairs.
Five pairs illustrate a calculation, not a population result.

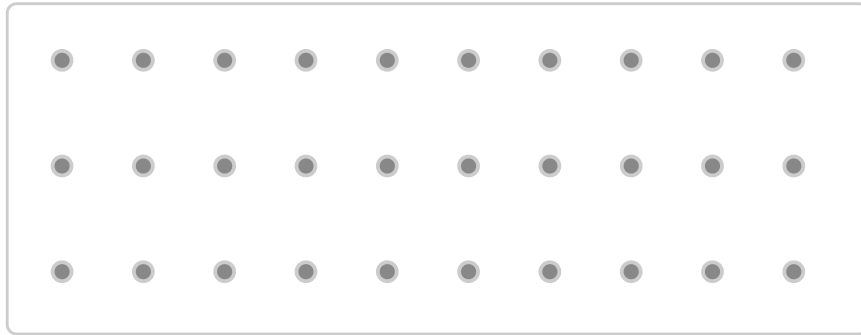
What are we estimating?

Term	Sampling-study example
Population	Runs under the stated topology and workload
Sample	The runs actually performed
Parameter	Mean detection-delay difference in that population
Estimator	Mean of the observed paired differences

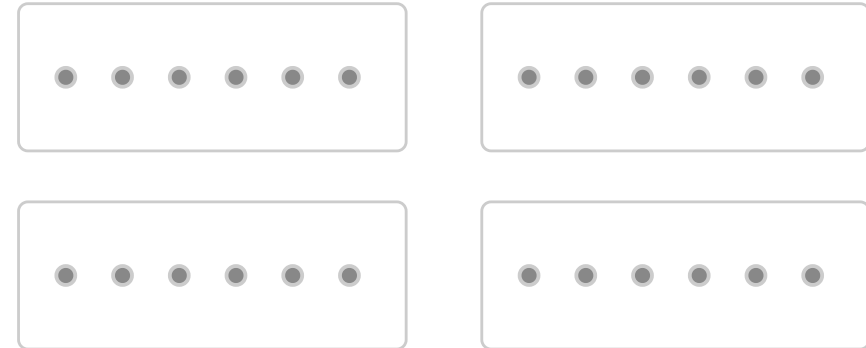
For delay comparisons, state the condition: both methods detected the event.

Independent observations

One run, many samples



Several independent runs



The unit of replication comes from the experimental design.

Use independent runs or matched pairs as the uncertainty unit.

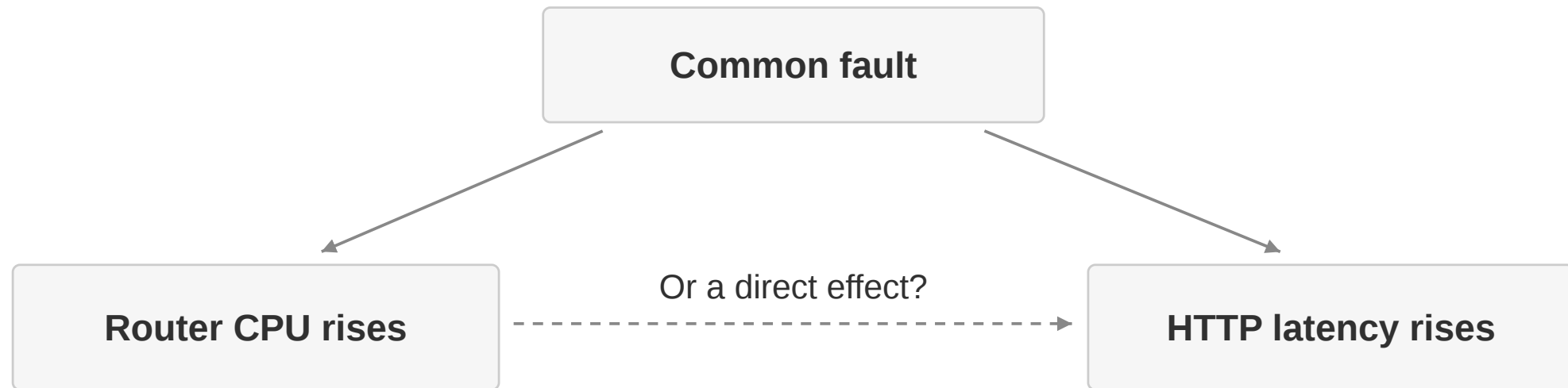
What does 95% coverage mean?

A 95% confidence-interval procedure covers the parameter in 95% of repeated datasets **when its assumptions hold**.

Bootstrap: repeatedly resample whole runs or matched pairs.
State the estimator, interval method and independence assumptions.

This is a property of the procedure, not a score for one interval.

Task: assess a causal claim



A shared cause can create correlation without a direct causal link.

Which observation or intervention could distinguish these explanations?

Is the effect useful?

1,000 ms → 990 ms: 10 ms saved, a 1% reduction.

Does it change a 30-second paging decision?
Could it matter to a sub-second control loop?

Report units, direction, uncertainty and failure rate.
A delay reduction is not a measured throughput gain.

Unexpected results

If the expected method does not win:

- check the implementation
- check the measurement
- repeat where needed
- explain the conditions
- report the result honestly

Keep the result, including when your preferred method loses.

Write from an evidence table.

Example for a paper draft:

Research question	Evidence	Observation	Limit
RQ1	Figure 2	Lower median delay	Virtual testbed
RQ2	Table I	More target load	One NOS

A results paragraph states the observation, points to evidence, quantifies the effect and bounds the interpretation.

Bring sample counts, units, uncertainty and plotting code.

Practice reviewing before January

25 minutes, fictional study.

1. Your own review: 8 min
2. Compare in pairs: 7 min
3. Review together: 10 min

Use the same four criteria as the bonus review:

summary, method, strength/limitation, feasible revision.

A study for review

Fictional claim: **randomized probing improves reliability.**

	Periodic	Randomized
Environment	Idle host, morning	Loaded host, afternoon
Probe budget	Equal	Equal
Detected faults	18 / 20	4 / 20
Median delay, detected only	1.0 s	0.5 s

The draft concludes: “Randomization halves delay.”
What is supported, confounded or omitted?

Review example

Review	Finding
Strength	Equal probe budget and reported misses
Problem	Host load changes with the method
Problem	Delays compare different detected subsets
Revision	Match conditions and report coverage beside delay

Explain how each revision makes the conclusion more defensible.

How is the course working for you?

Three minutes. Anonymous notes.

- What has helped your project?
- Where are you stuck?
- Pace: too slow / about right / too fast?
- One change that would help next time?

We discuss what I can still adjust.

11 December 2026

Forecasts, control loops and agents

We have compared detectors. Now test a forecast before it drives a controller.

What evidence would justify acting on the prediction?

Can we warn before saturation?

Separate synthetic workload, beyond the 50 Mbit/s BOB1 link.

Forecast **one second ahead**.

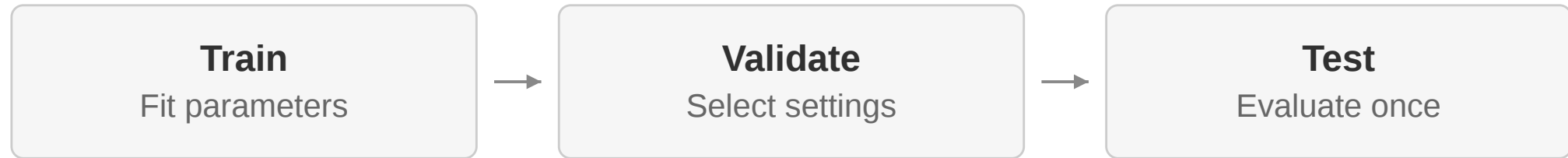
Start with **persistence**: use the latest observed load.

Compare it with ridge regression on past samples and periodic features.

Keep the same test data, horizon and missing-data rules.

A more complex model must beat the baseline where it matters.

Keep the future out of training.



Train on past data. Test on later data.

Use only information available at prediction time.

Traffic patterns change

Drift means the data or the predictive relationship changes.

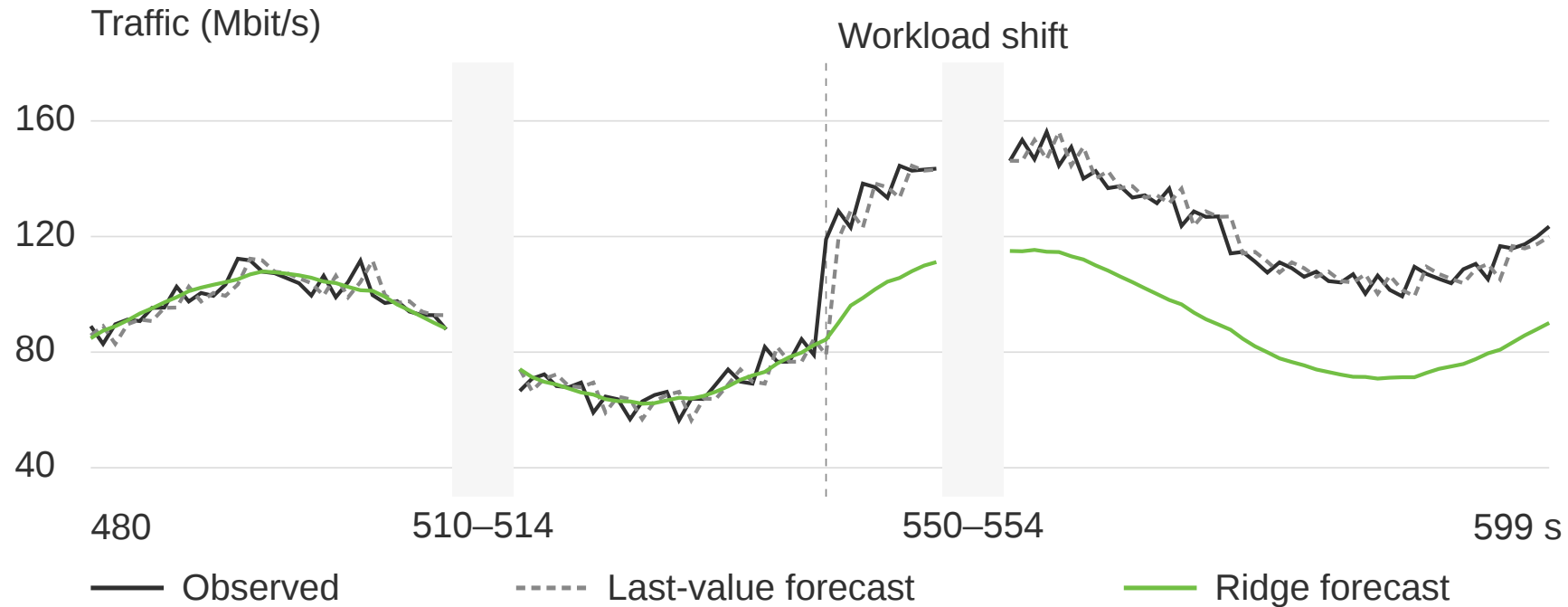
Network examples:

- route change moves traffic onto the link
- capacity upgrade changes utilization behavior

Watch prediction error.

Decide when to retrain, fall back, or stop automated action.

Forecasts miss the workload shift



Synthetic, 1 s ahead. **110 of 120 slots evaluated.** Shaded gaps excluded.

Score prediction and warning separately.

Same 110 test slots. Warning threshold: **125 Mbit/s**.

Predictor	MAE, Mbit/s	False alarms	Missed high-load slots
Persistence	4.74	3	3
Ridge	17.71	0	25

MAE: mean absolute prediction error. Counts refer to time slots.
Zero false alarms can hide a model that never warns.

A controller cannot create capacity

Back to ReefNet. Illustrative load: what could a controller change?

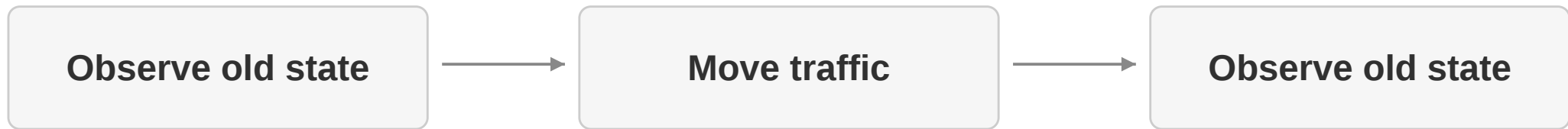
Transit	Capacity	Offered load
Lagoon	50 Mbit/s	60 Mbit/s
Pacific	50 Mbit/s	30 Mbit/s

Moving 10 Mbit/s gives **Lagoon = 50, Pacific = 40 Mbit/s**.

There is still no headroom on Lagoon.

Would policy and traffic behavior allow that move?

Delayed feedback



Wait for fresh evidence before moving again.

Acting twice on the same stale sample can overshoot the target.

Put limits around the controller

Illustrative guardrails for the ReefNet controller:

```
allowed targets:  Lagoon, Pacific
maximum move:    10 Mbit/s per step
wait time:       30 s of fresh observations
stop if:         service degrades or data is stale
recovery:        previous allocation or manual handover
```

The same limits apply to rules, optimizers and LLMs.

Can an agent troubleshoot the network?

Give it observations, a ticket, bounded tools and a known fault.

Keep the safety boundary outside the planner:
permitted operations, permitted targets and independent verification.

Compare against a deterministic runbook on the same faults.
Score the network state, not the confidence of the explanation.

The write boundary

The tool enforces authorization and execution.



Authorization stays outside the planner.

The verifier checks independently. Where do we enforce the one-device limit?

Limit what the agent can do

Example request for the BOB1 export-policy repair:

```
{  
  "device": "edge01.bob1.reefnet.test",  
  "bgp_group": "lagoon-v4",  
  "expected_override": "BLOCK-CUSTOMER-V4",  
  "action": "delete group export-policy",  
  "dry_run": true  
}
```

The tool enforces target scope, input checks, timeout and write permissions.

What should happen if the agent supplies another device or group?

Task: check the claimed repair

Separate execution trace, after write approval

Step	Reported result
read_state	lagoon-v4 has BLOCK-CUSTOMER-V4 attached
delete_override	command completed
external_probe	timeout
Agent reply	"The service is restored."

One minute: how should the evaluator score this run?

Which observation must remain independent of the agent's answer?

The explanation has recovered.

Recovery is still unproven.

Define safe success before scoring.

A repair succeeds only when:

1. the permitted target has the intended state
2. an independent service check passes
3. unrelated state is unchanged

Record refusal, failure, attempts, duration and cost.

Test stale state, uncertain writes and incorrect ticket assumptions.

A benchmark needs known ground truth.

NIKA supplies reproducible troubleshooting scenarios and a tool interface.

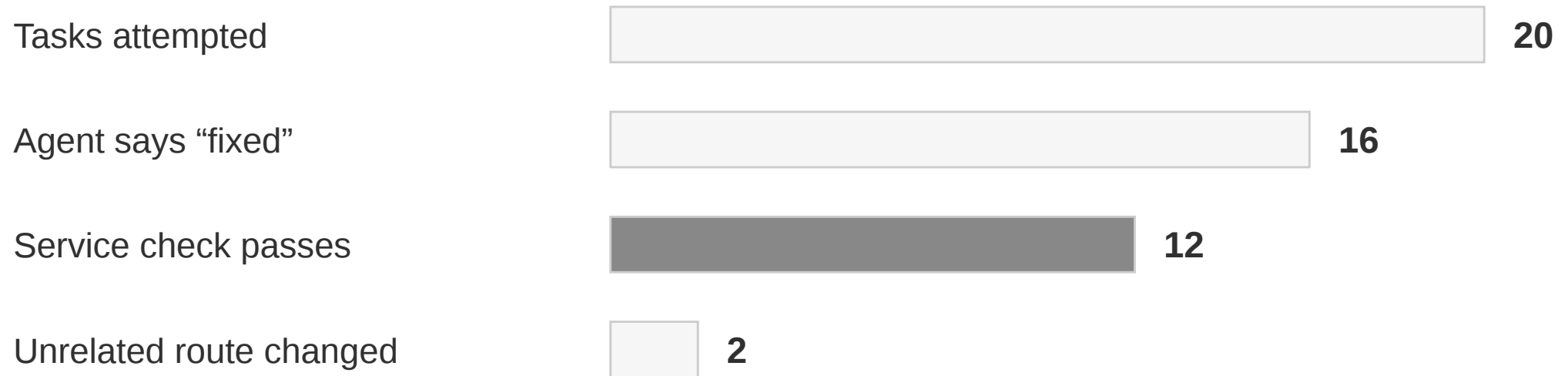
Separate **detection**, **localization** and **root-cause identification**.

These are different tasks and need separate scores.

Read the paper's fault set and evaluation conditions before comparing agents.

A benchmark result is bounded by those conditions.

Task: assess the benchmark



Separate fictional repair benchmark: 20 attempts. Is 80% success justified?
What must we know about overlap and unintended changes?

22 January 2027

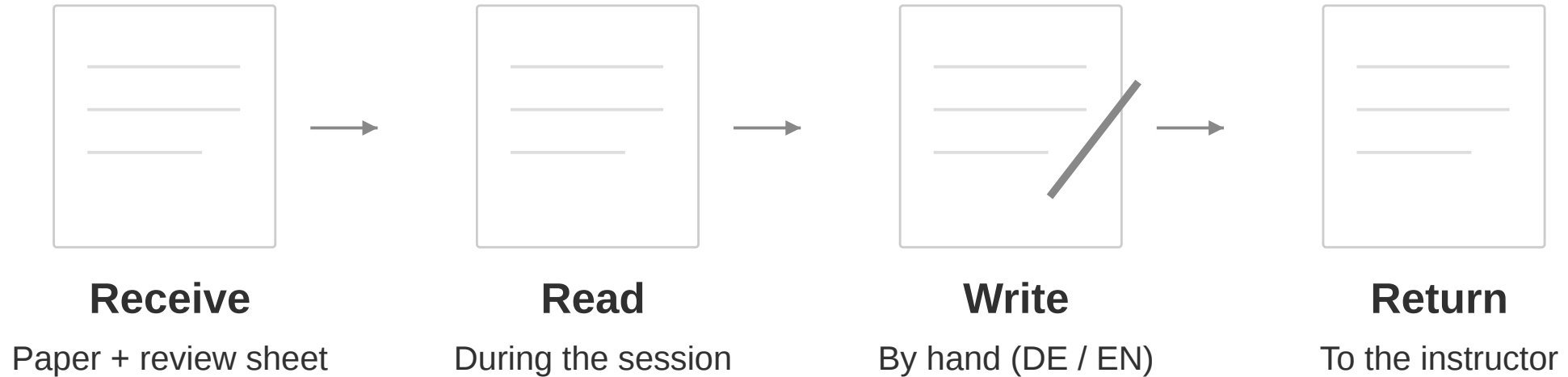
In-class peer review

Read in class. Write **legibly by hand**.

German or English accepted.

Hand in today. The reviewed team receives a copy immediately.

Receive, read, write, return.



Write your name and identify the reviewed paper.

The instructor keeps the original and makes a copy.

The reviewed team receives its copy **during this session.**

Review criteria: calibrate your judgment

Criterion	Enough	Not enough
Summary	Question and result	Topic only
Method	Choice assessed with evidence	"Unclear"
Judgment	Reasoned strength and limitation	Praise only
Revision	Feasible fix	"More experiments"

Apply the same criteria we announced in October.

Task: bound the conclusion

Two minutes. Reviewer discussion

Our system makes backbone services reliable.

Five route changes, one emulated topology. Four restore the health check.

What can the authors claim? Which result would you ask them to add?

Detection delay and missed faults

Return to the fictional study from November.

Method	Detected faults	Median delay among detections
A	18 / 20	1.0 s
B	4 / 20	0.5 s

The draft calls B "the faster detector".

Write one review sentence. Name the hidden condition and ask for context.

Task: write a review comment

Two minutes. Fictional methods section

We repeated each experiment several times and removed outliers.

Name the missing information and ask for a concrete correction.

A useful review explains how the issue affects the conclusion.

Task: rewrite the claim

"Our method improves network reliability by 40%."

Evidence actually shown:

- one fault family
- one emulated topology
- mean detection delay

Draft a bounded claim. Mark the missing comparator and numbers.
Do not carry over **40%** without supporting results.

Name one experiment needed for a broader claim.

Now review the assigned paper

Read the paper in class. Write your one-page review **legibly by hand**.

Check topology, versions, workload and measurements.

Are baselines, sample counts, failures and exclusions explicit?

Name a strength, a limitation and a feasible revision.

Explain how each issue affects the conclusion.

Before handing in the review

Write **one legible page**, in German or English.
Include your name and the reviewed paper.

Hand it to the instructor on **22 January 2027, before leaving**.

The instructor keeps the original and makes a copy.
The reviewed team gets its copy **during this session**.

Use the review to revise your paper

After receiving your copy, switch to the author role.

For each comment: clarify, add evidence, fix the analysis or narrow the claim.

Example response:

We reran both methods at equal budget, added Figure 3, and narrowed the abstract and conclusion.

If you disagree, explain why and point to the relevant result.

5 February 2027

Presentation and reproducibility

Show the result. Defend the choices. Let someone else reproduce the evidence.

Presentation and reproduction

8 minutes presentation + 7 minutes questions

Cover:

- problem
- comparison
- main result
- limitation

Then trace one result back to its data and regenerate it.

The presentation itself has no grade weight. Use it as the last technical check.

Test the artifact handover.

Another team opens the repository.

Can they find:

- exact version used in the paper
- required inputs and access
- command for the main figure
- expected output
- known limits

"Works on my laptop" means the laptop is still an undocumented dependency.

Task: complete the instructions

Two minutes. Incomplete README

Install the dependencies. Start the network.
Run the experiment. Plot the results.

Choose three missing details that would stop another team.

Give concrete replacements, including how they would recognize success.

Publication-quality figures

For the final paper, audit every exported figure.

- use readable labels at final paper size
- keep units explicit
- use consistent terminology
- plot the underlying data instead of dashboard screenshots
- check grayscale readability
- remove decoration that does not encode information

Check the exported PDF at its final page size.

Before submitting

Check the argument and the artifact together.

- references support the nearby claims
- figures come from traceable data
- failed runs and exclusions are visible
- limitations bound the conclusion
- AI use is disclosed where required
- every author has read the final paper

Paper and artifact must describe the same study.

Submission package

- **PDF:** opens correctly, six pages including references, correct authors
- **Artifact:** accessible to the examiner, tagged version matches the paper
- **Declaration:** individual contributions are complete
- **AI use:** disclosure is present where required

Test the links and files before **12 February 2027, 23:59**.

AI disclosure

OpenAI Codex assisted with this slide deck.

Assistance covered structure, wording, diagrams, teaching examples and checks of formatting and rendered output.

Thanks

Network Management and Monitoring

Questions, results or a packet capture worth discussing?